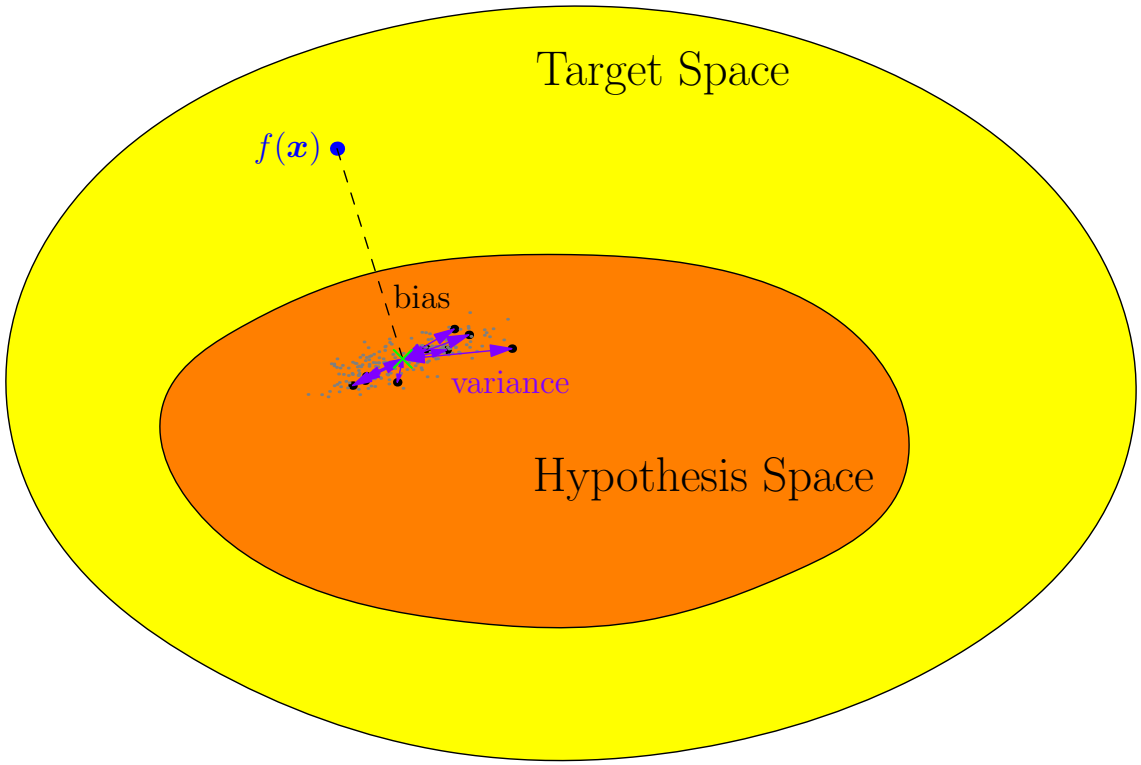
Mathematics of Machine Learning

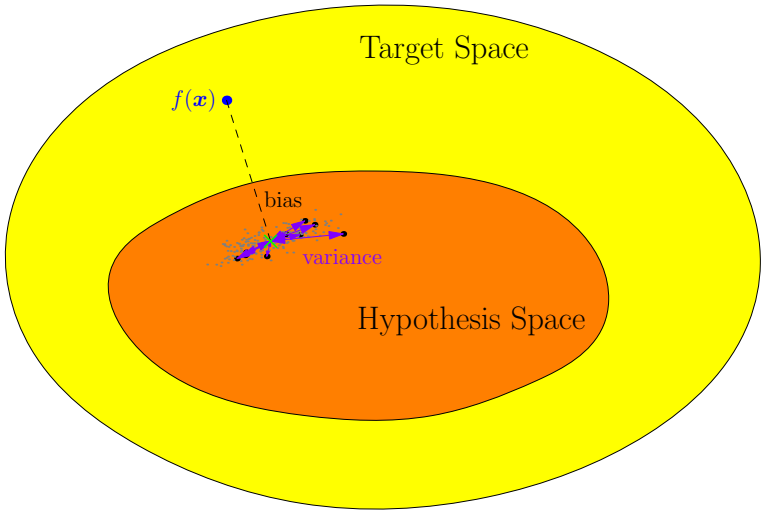
When Machine Learning Works



When ML Works, Bias Variance

Outline

1. **What Makes a Good Learning Machine?**
2. Bias-Variance Dilemma
3. Shrinkage and Ridge Regression
4. Ensembles



What Makes a Good Learning Machine?

- We want to understand why some machine learning techniques work well and others don't
- To understand why they work we need to understand what makes a good learning machine
- For this we have to get conceptual and think about **generalisation** performance

  ***generalisation**: how well do we do on unseen data as opposed to the training data*

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- The problem can be over-constrained (i.e. we have conflicting data to deal with)
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Least Squares

- Suppose we want to learn some output y for a feature vector $\boldsymbol{x}$
- We construct a learning machine that makes a prediction $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta})$
- We typically choose the machine to minimise a *training loss*

$$L_T(\mathcal{D}) = \sum_{(\boldsymbol{x},y)\in\mathcal{D}} \left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}) - y\right)^2$$

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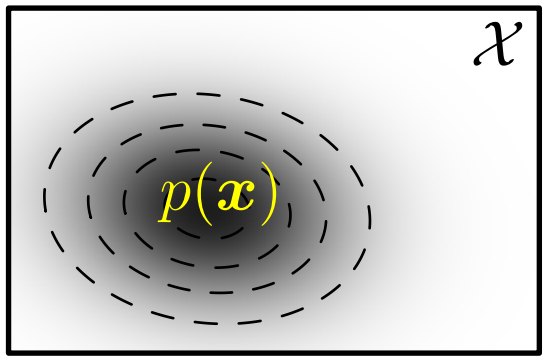
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- We call this machine $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})$

Generalisation Error

- We want to minimise the *generalisation loss* which in this case is

$$L_G(\boldsymbol{\theta}_{\mathcal{D}}) = \sum_{(\boldsymbol{x},y)\in\mathcal{Z}} \mu(\boldsymbol{x},y)\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - y\right)^2$$



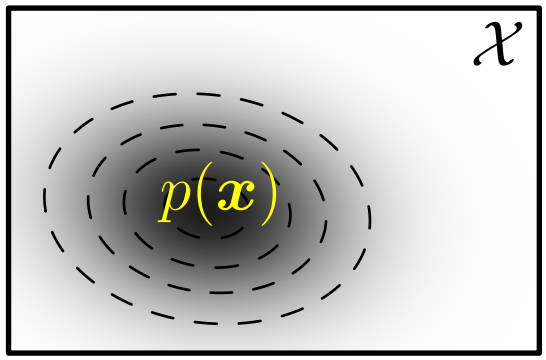
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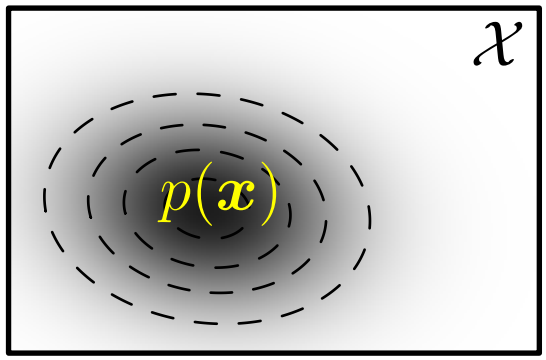
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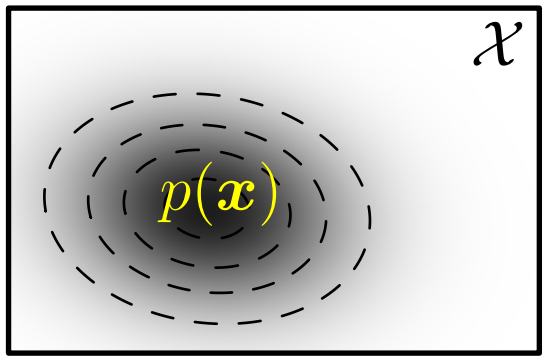
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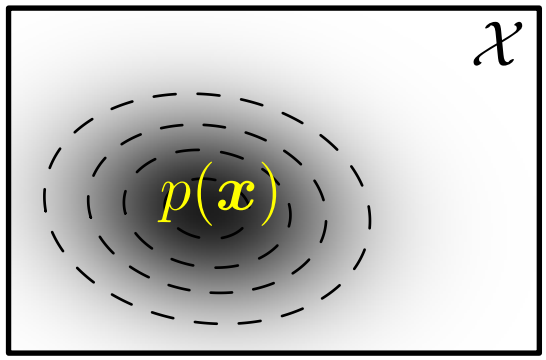
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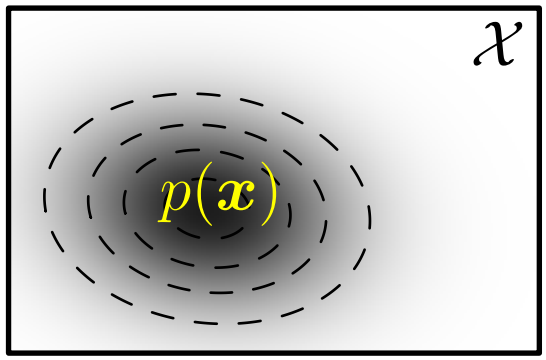
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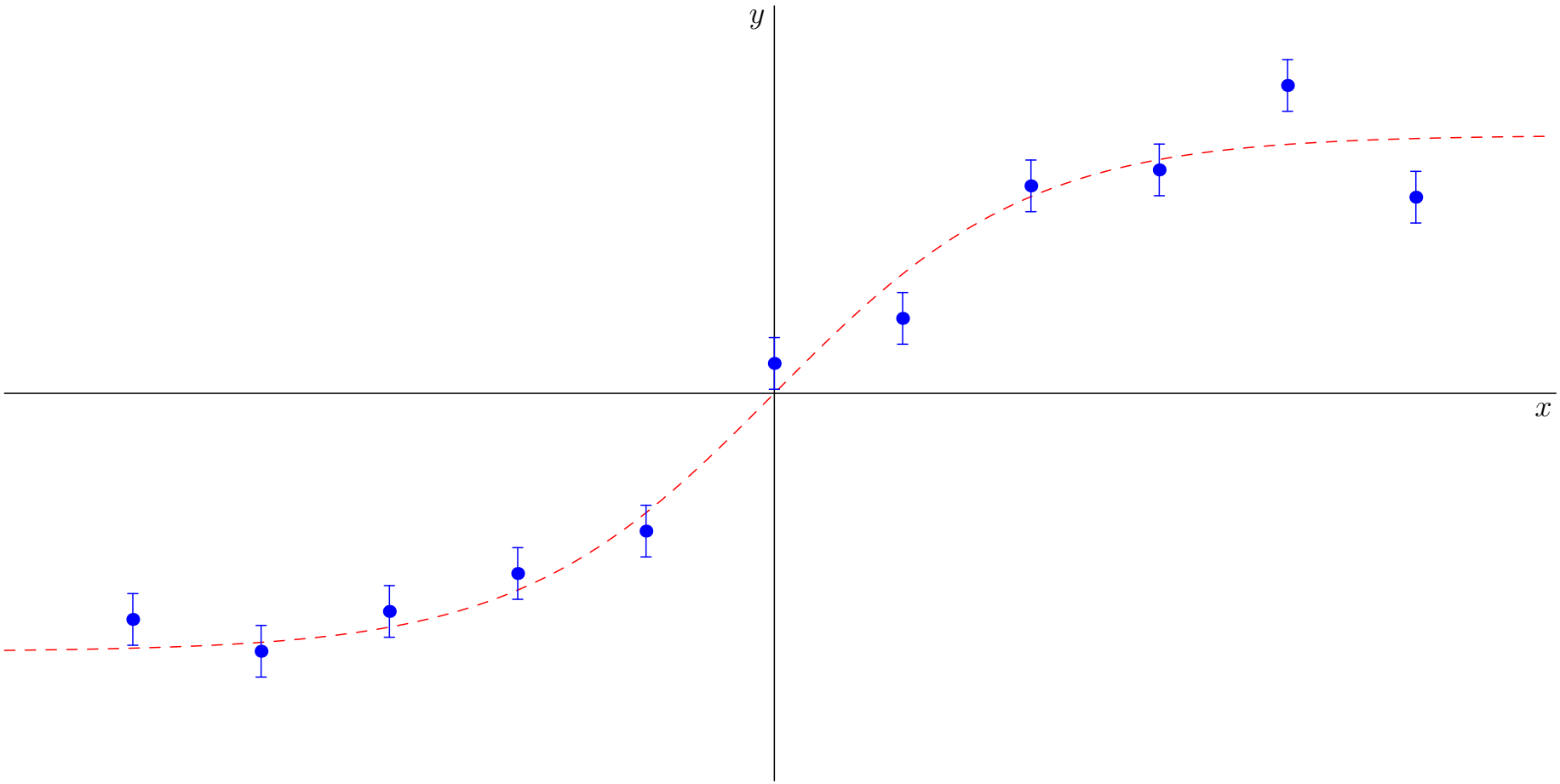
- We want to minimise $L_G(\boldsymbol{\theta}_{\mathcal{D}})$ but in practice we areb minimising $L_T(\mathcal{D})$, *what could possibly go wrong?*

Too Simple or Too Complex?

- Fit $\hat{f}(x|\boldsymbol{\theta}_{\mathcal{D}})$ to data

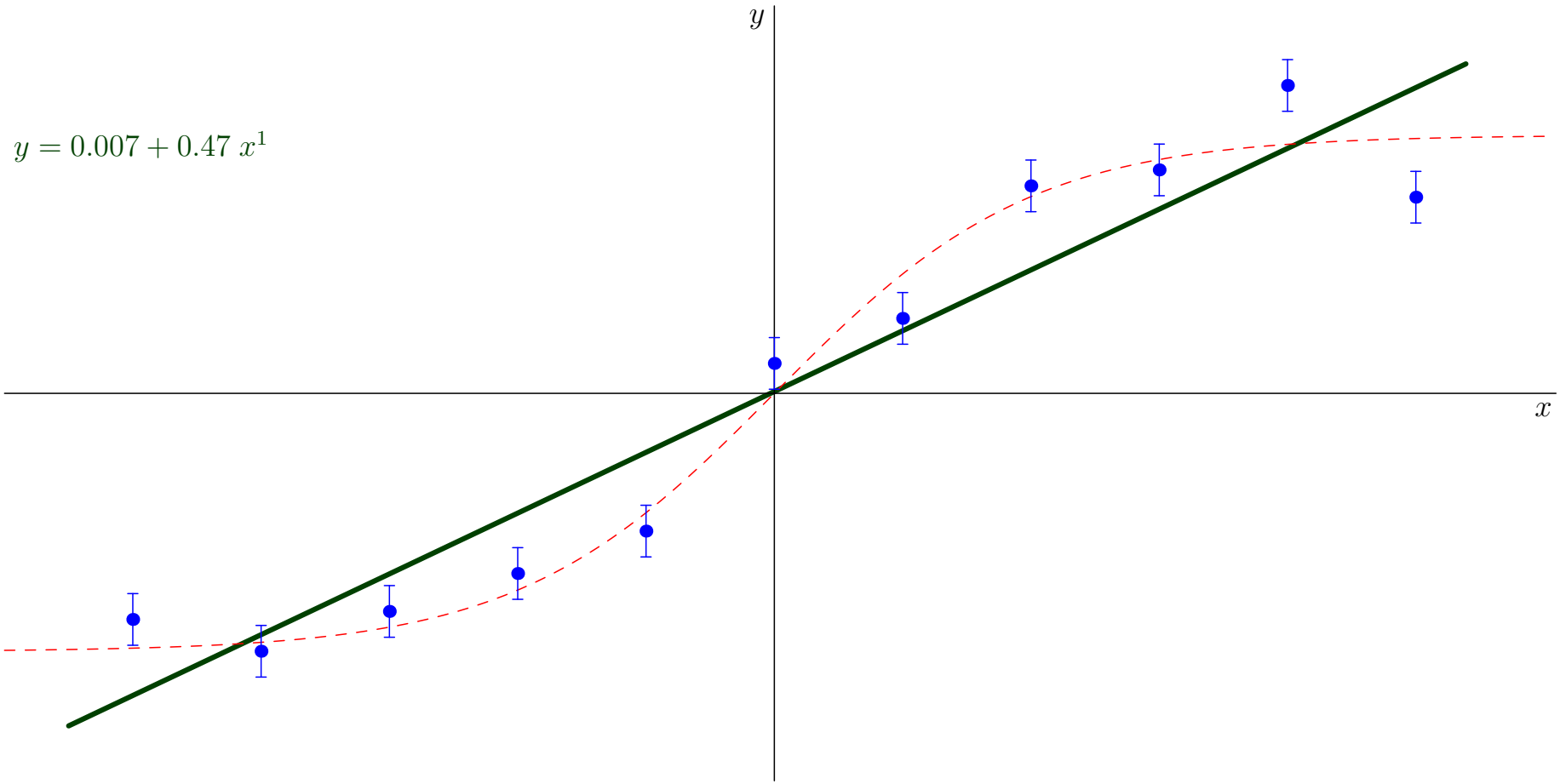
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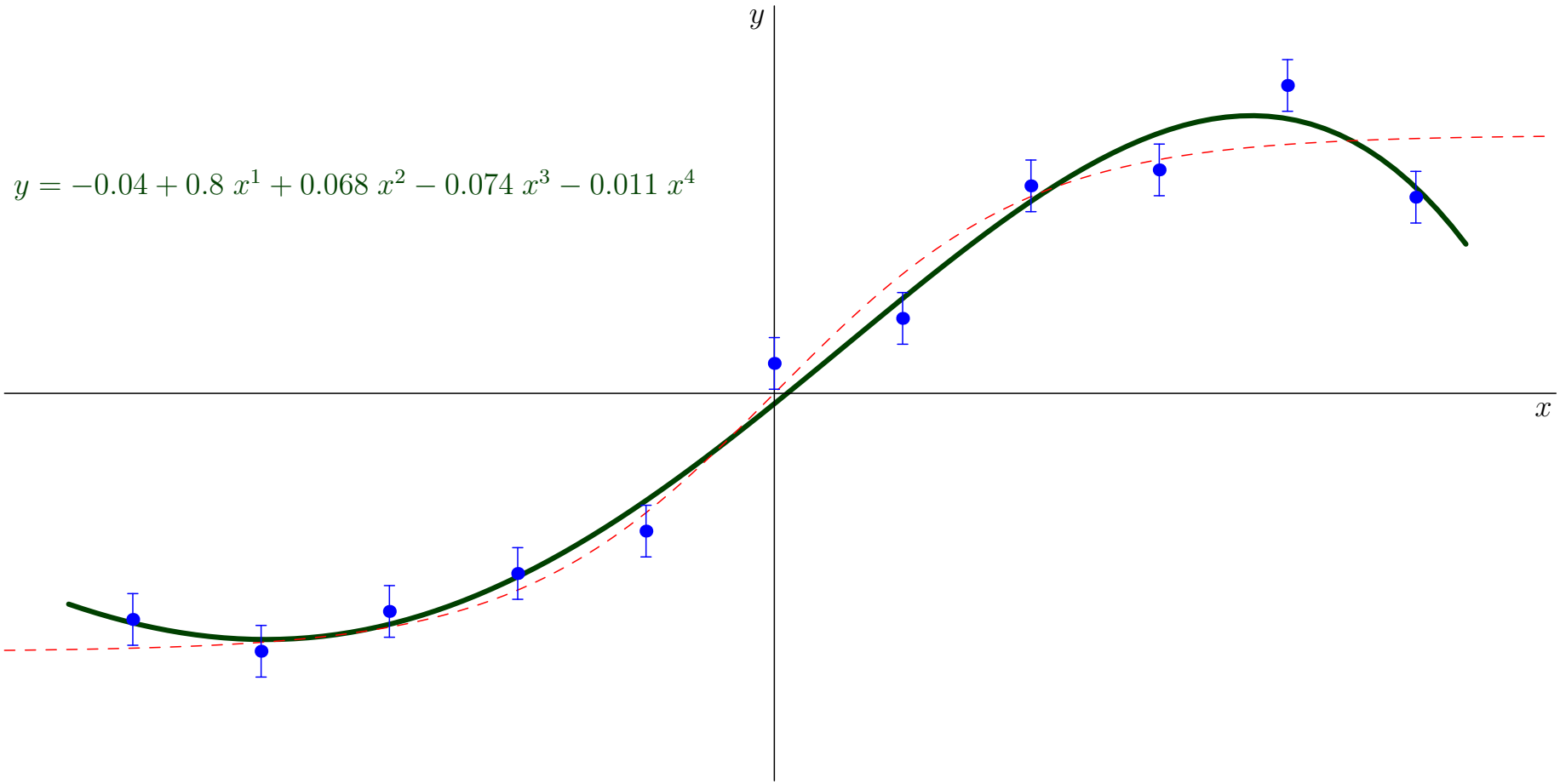
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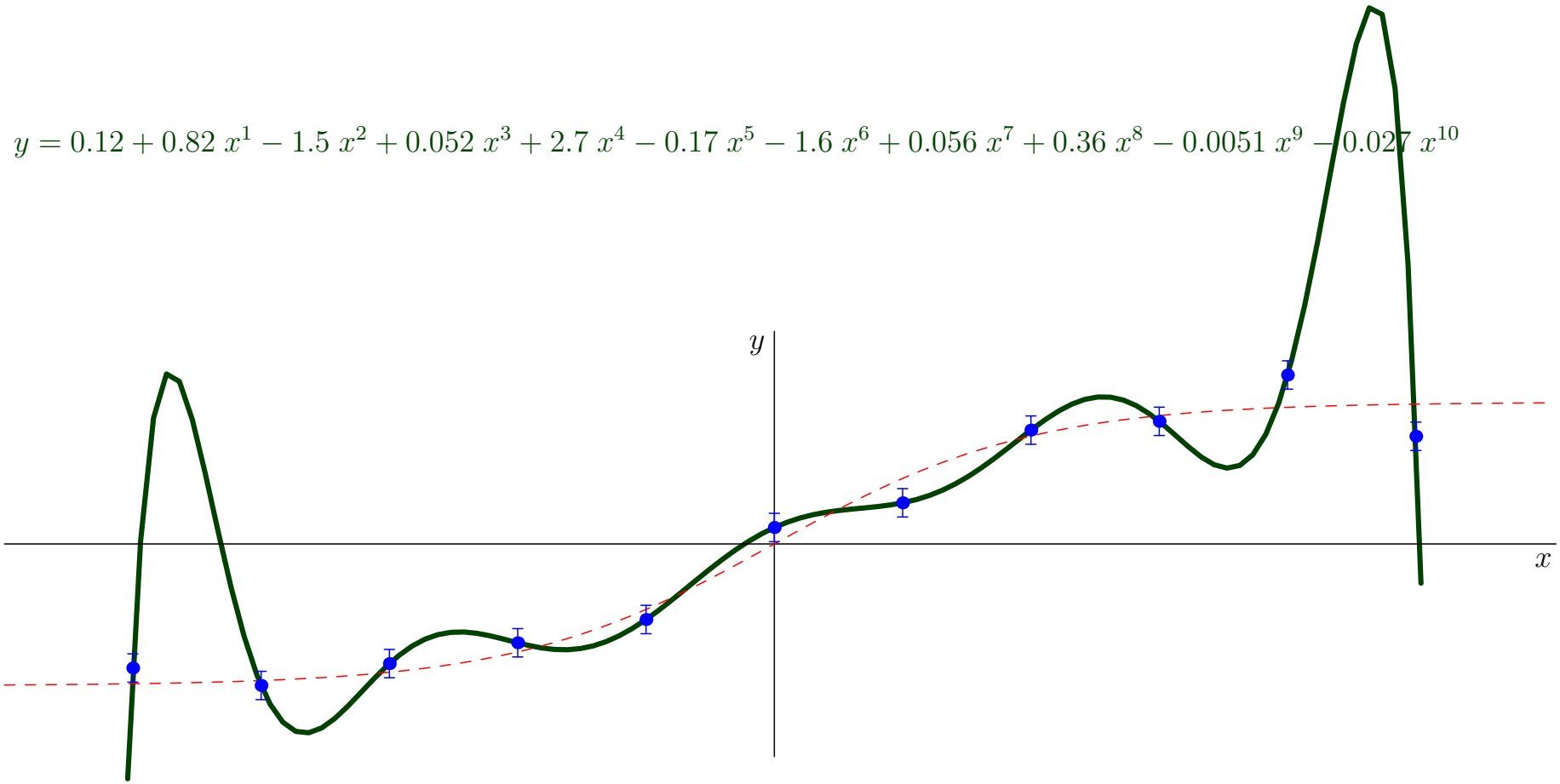
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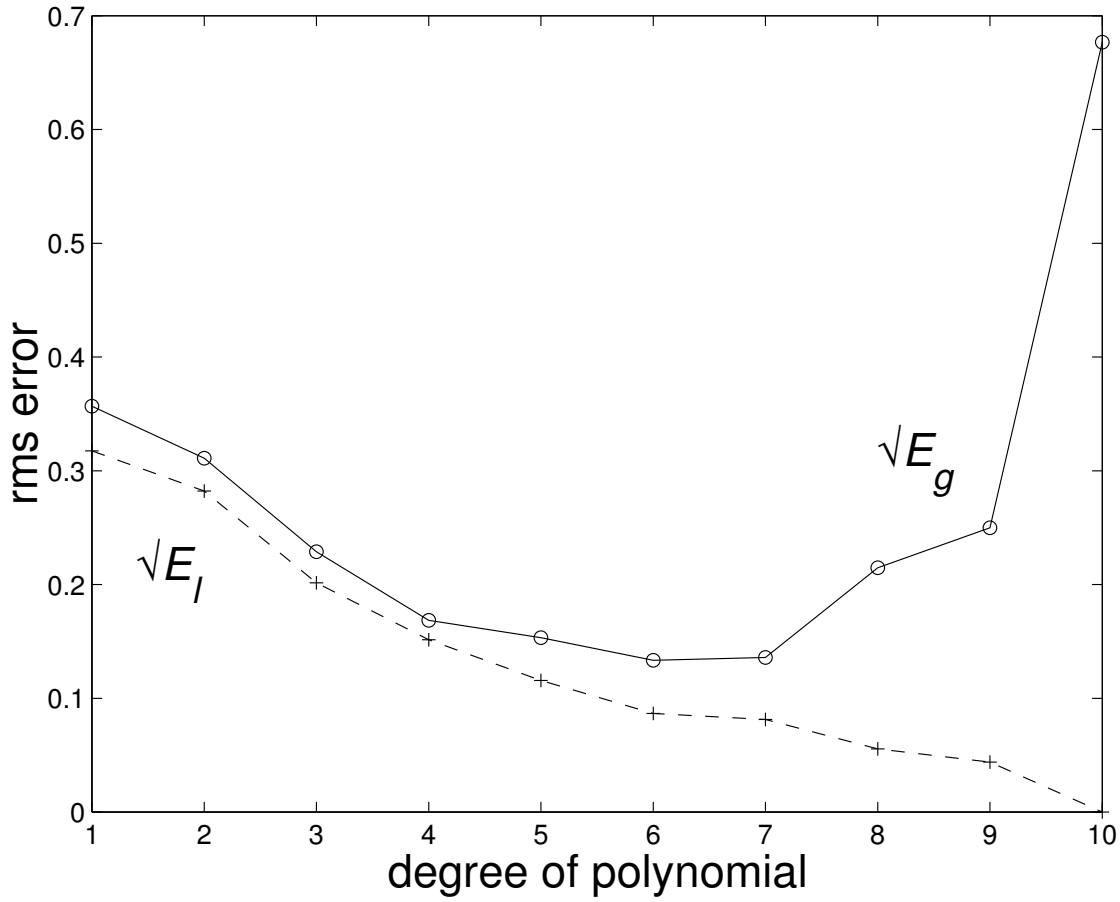
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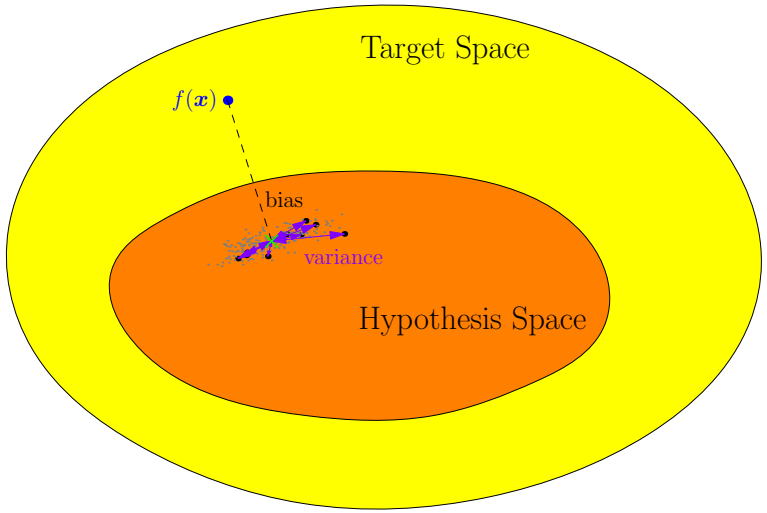
Measuring Generalisation Error for Regression

- Consider the regression example. The root mean squared error is



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- To reason about generalisation we can ask what is the $expected\ generalisation\ loss$, when we average over all different data sets of size m drawn independently from $\mu(\boldsymbol{x},y)$
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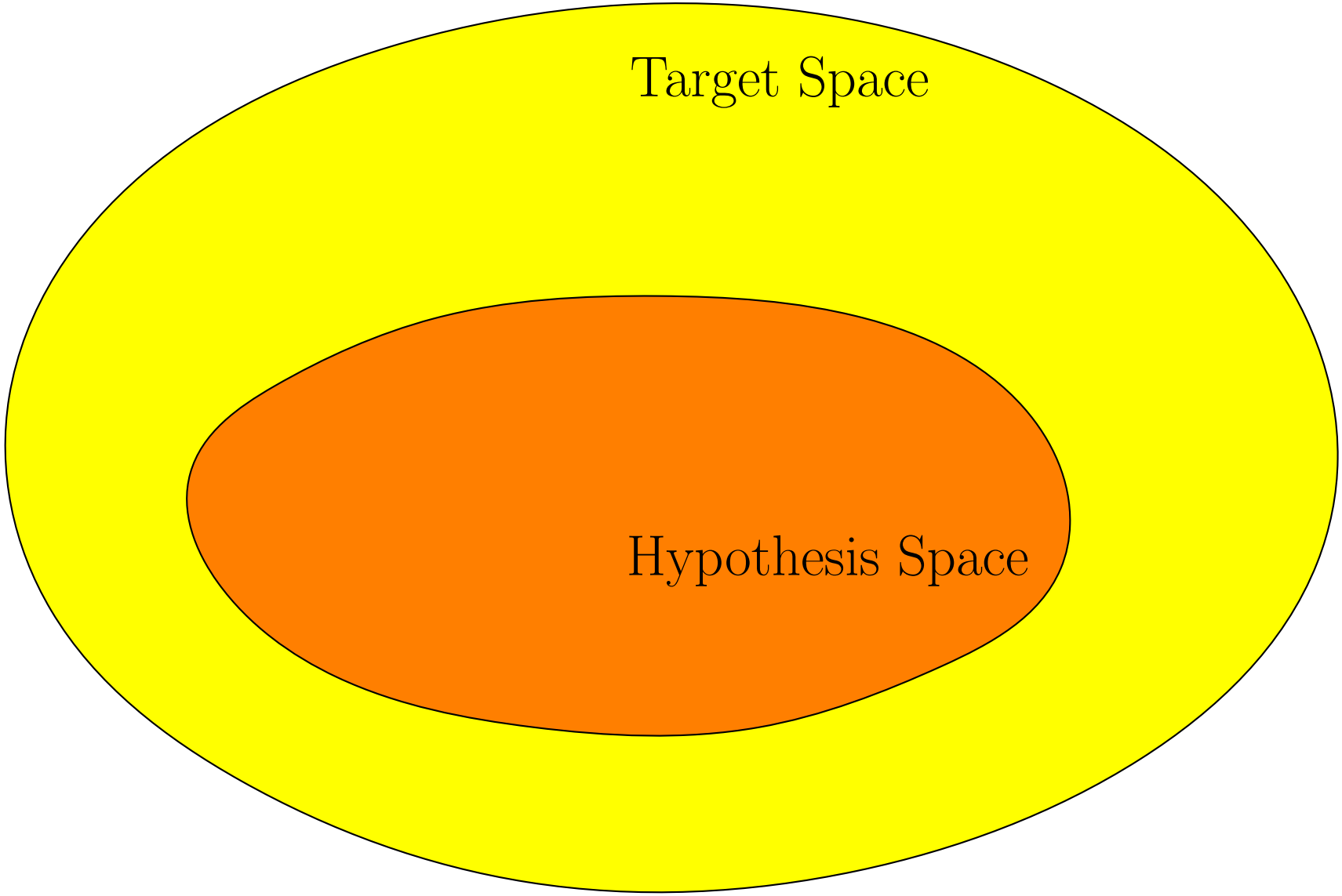
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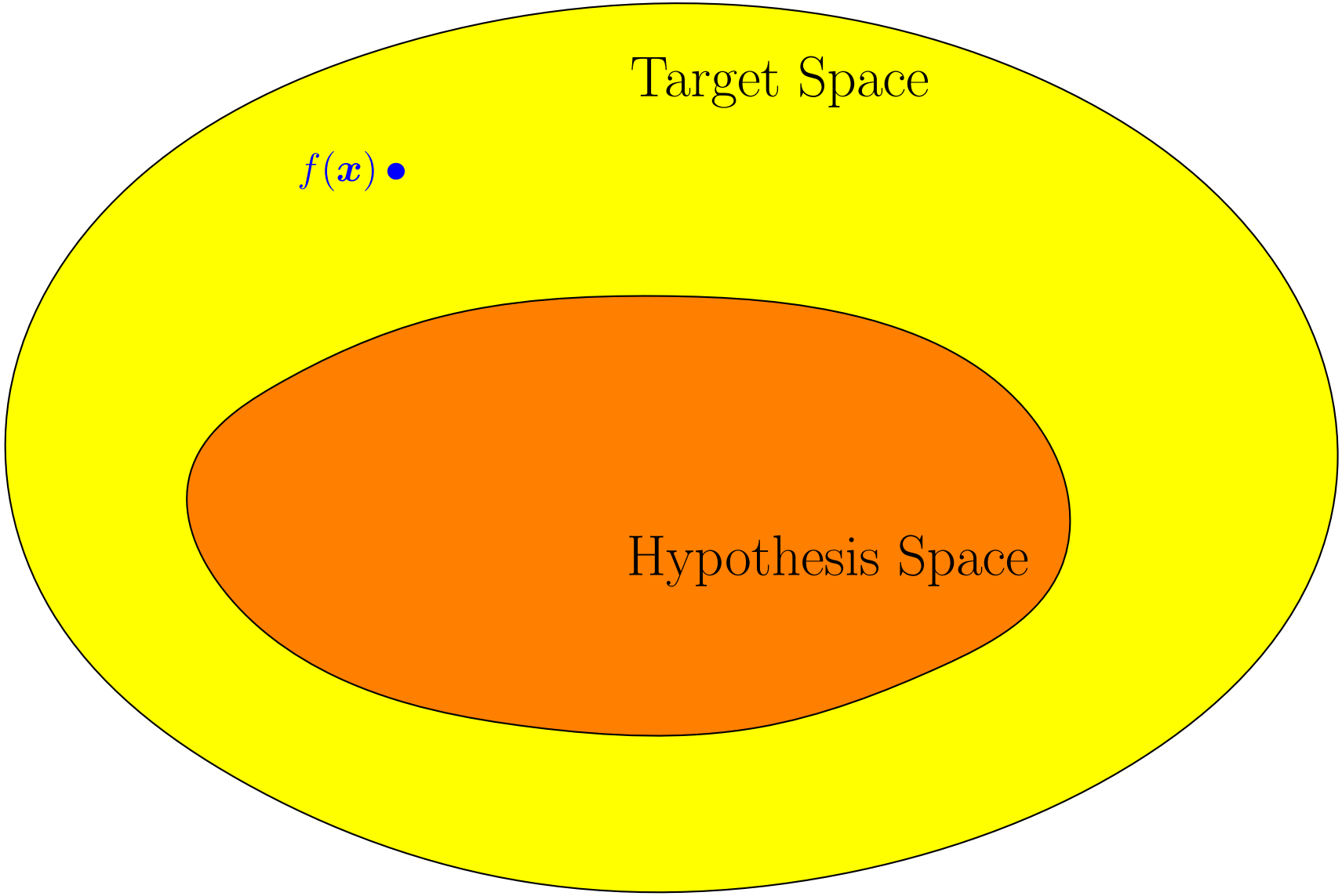
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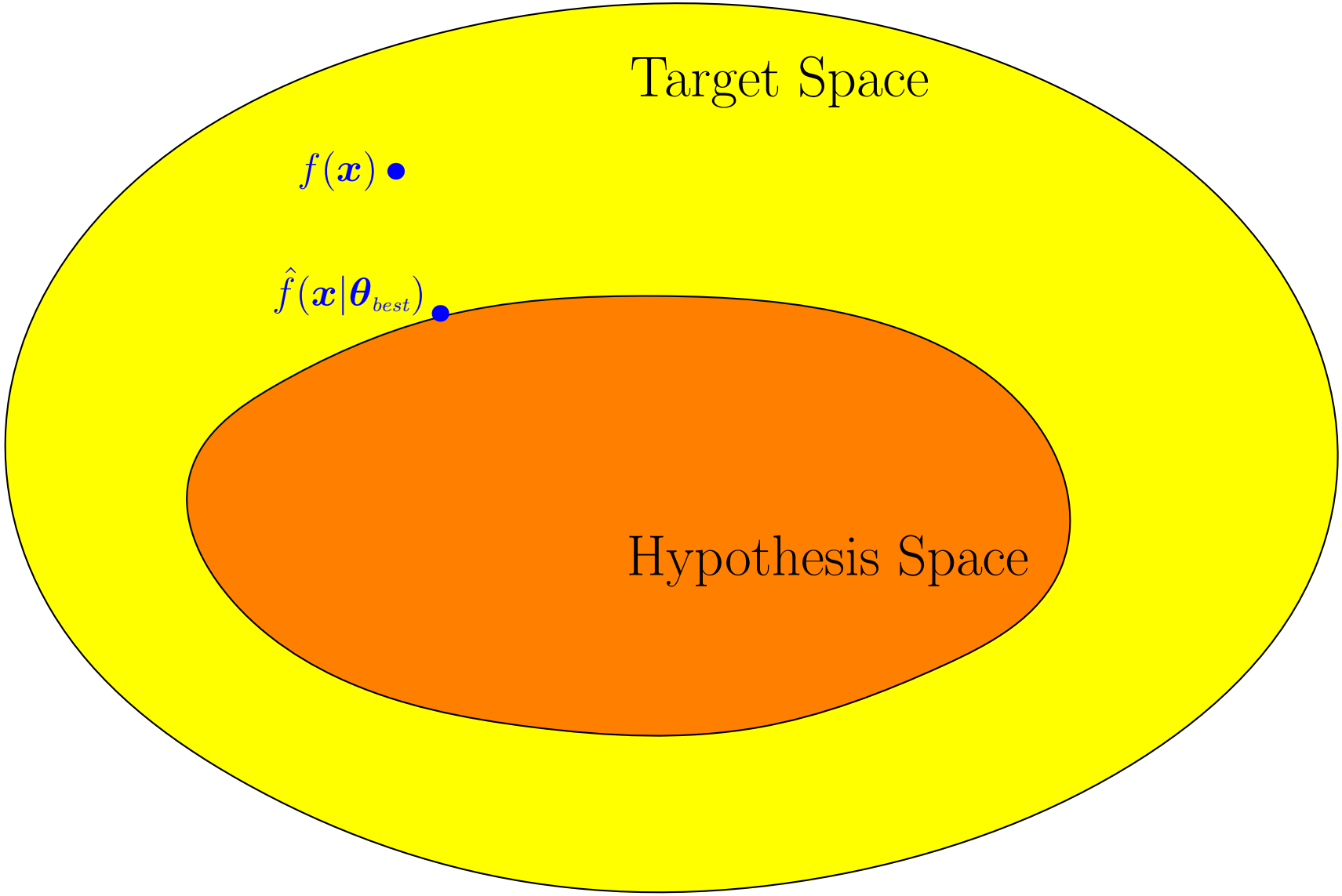
Approximation and Estimation Errors



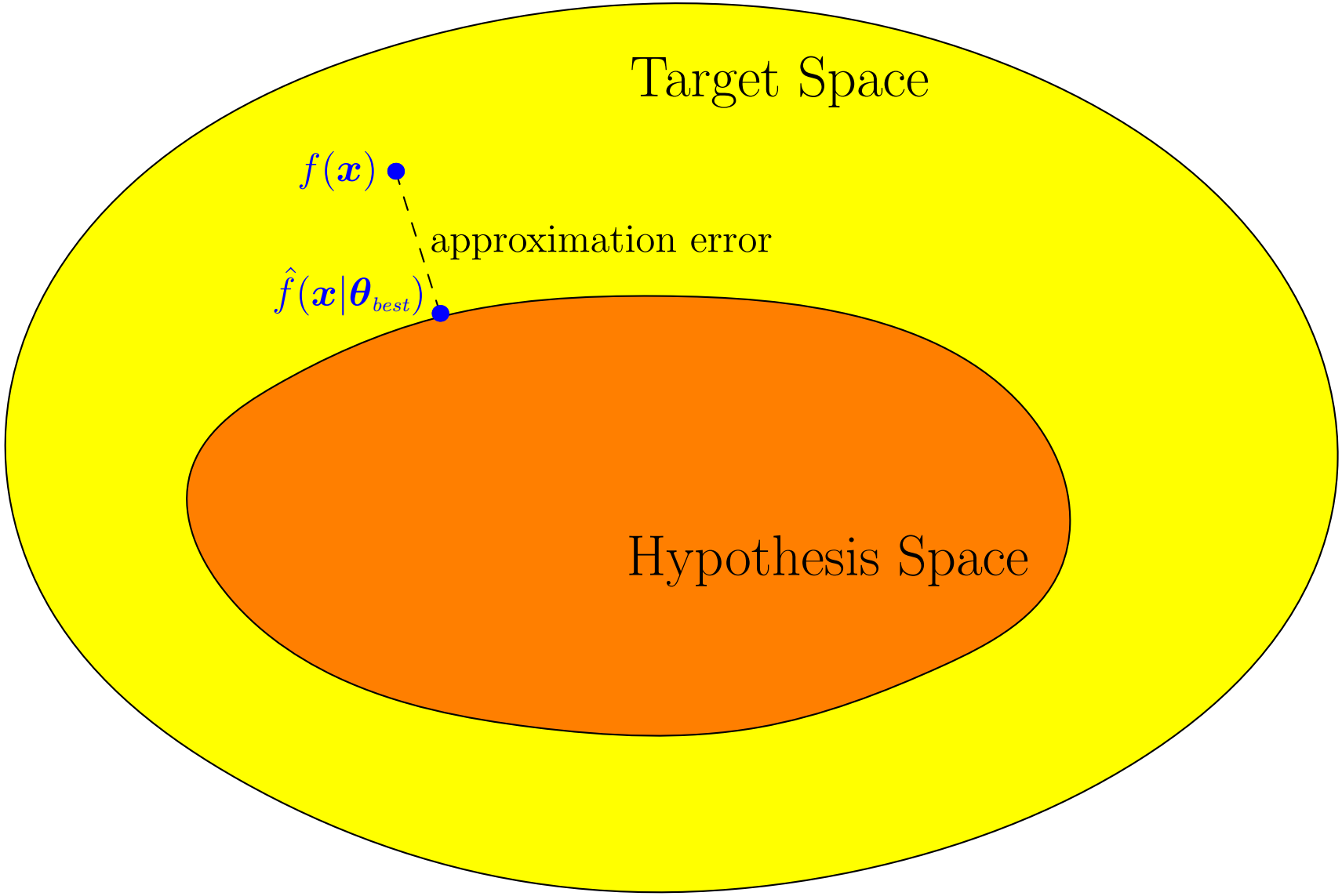
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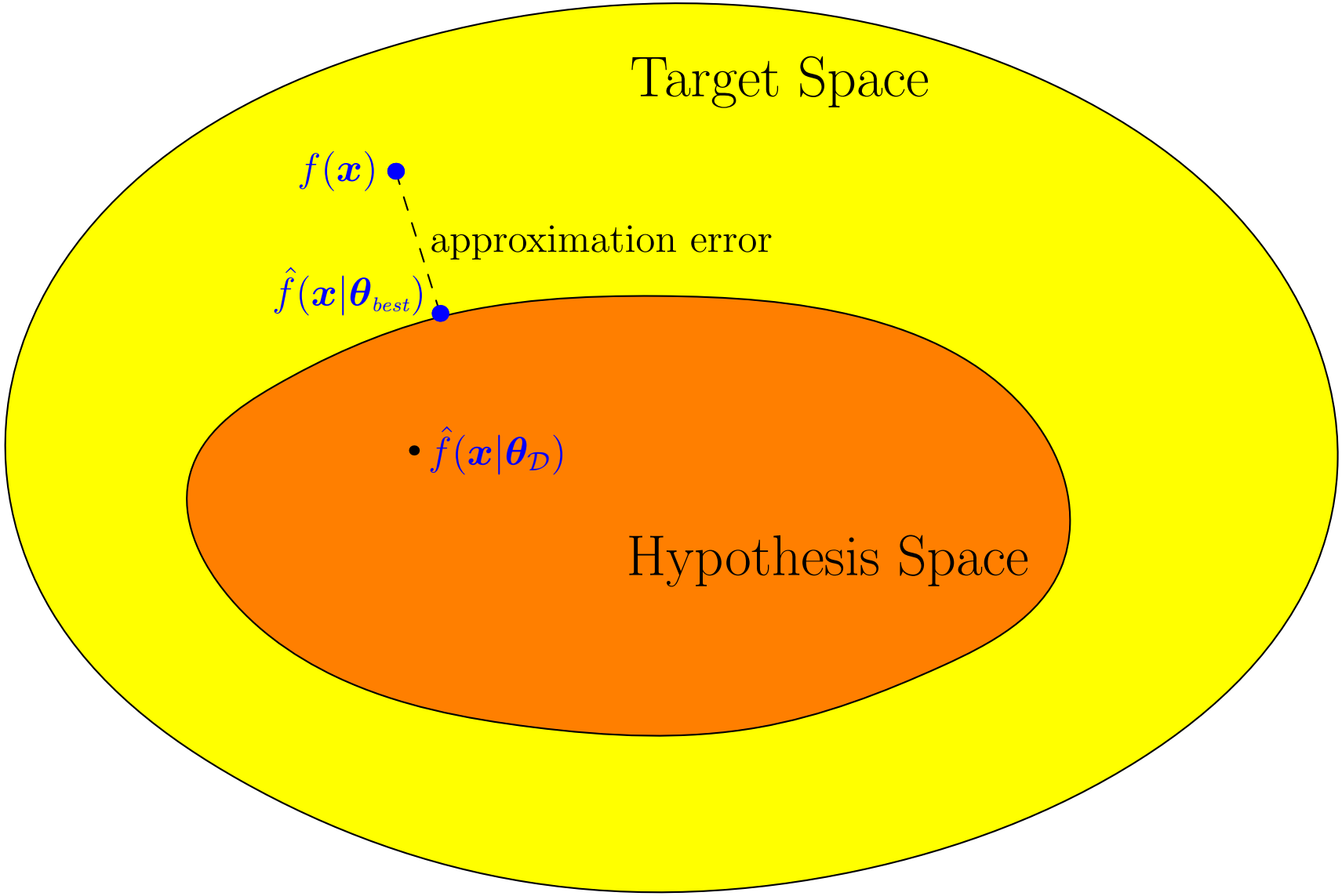
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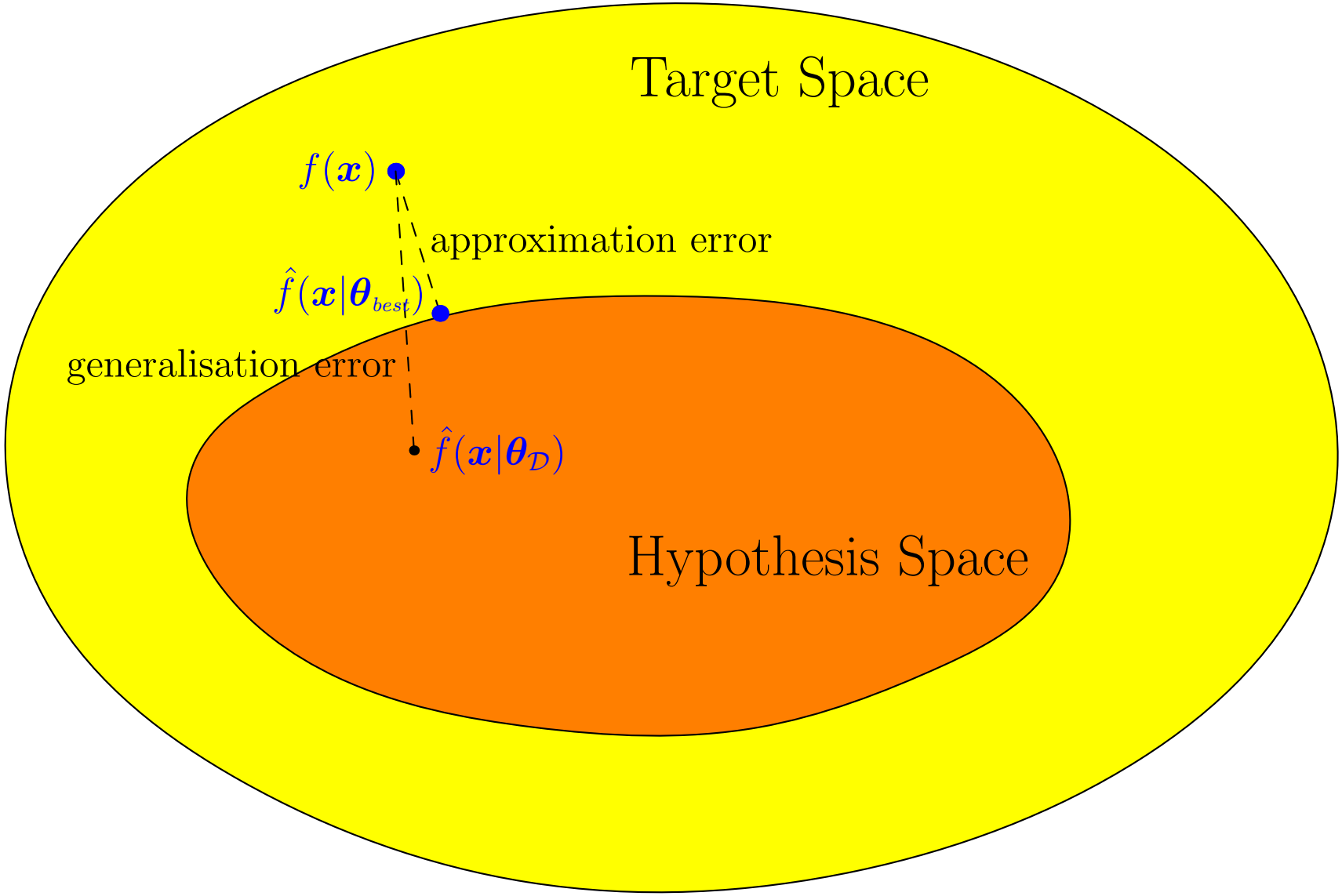
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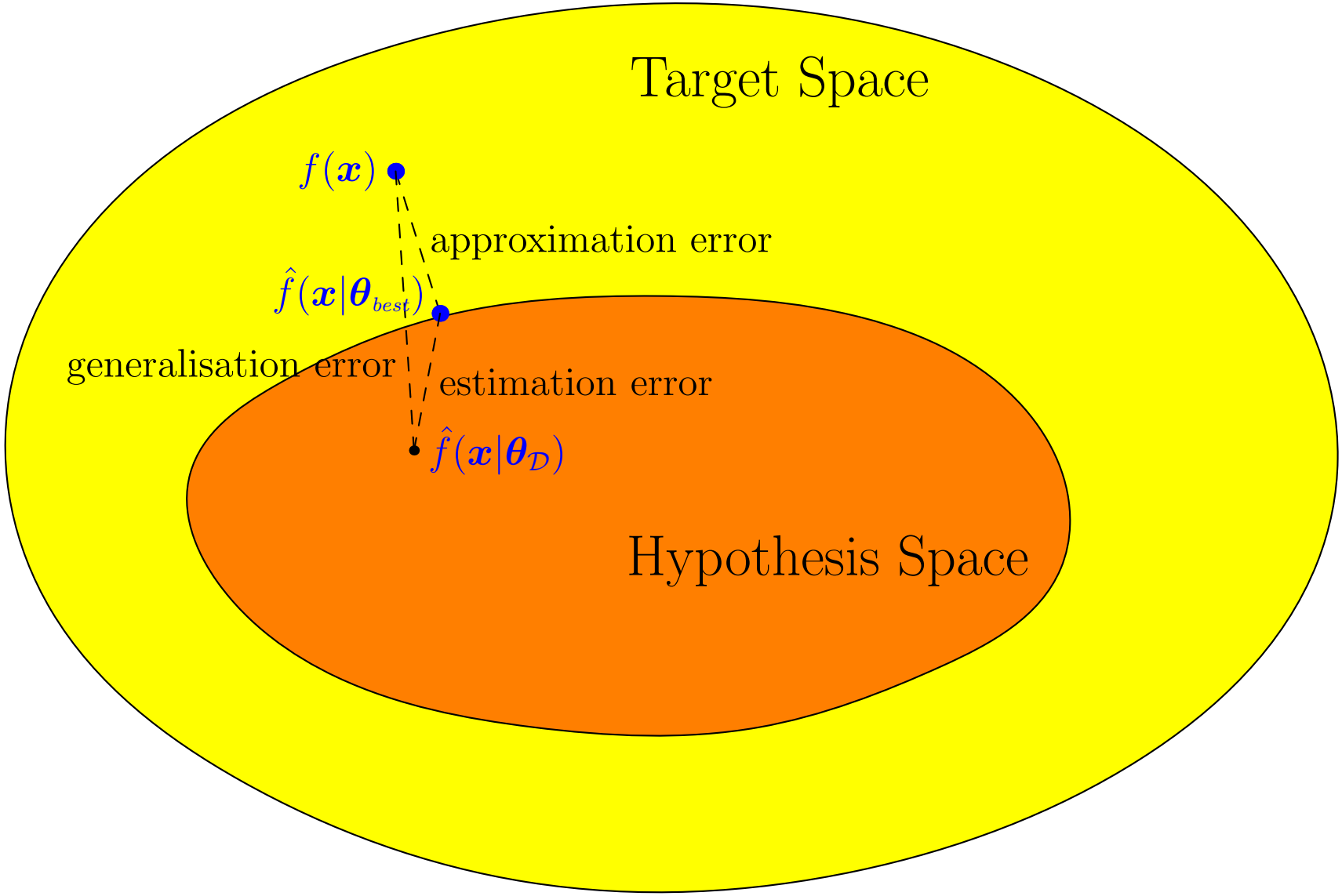
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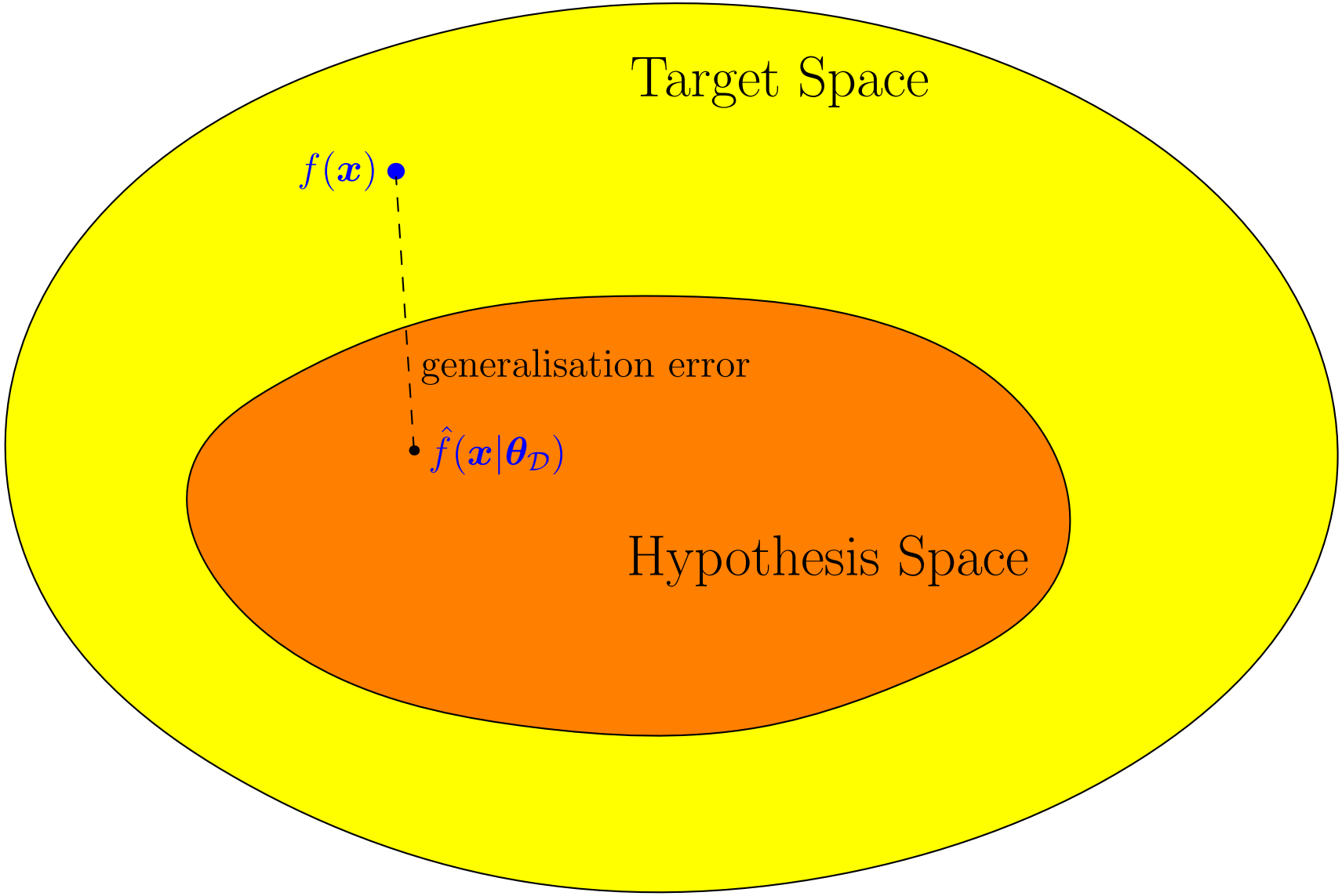
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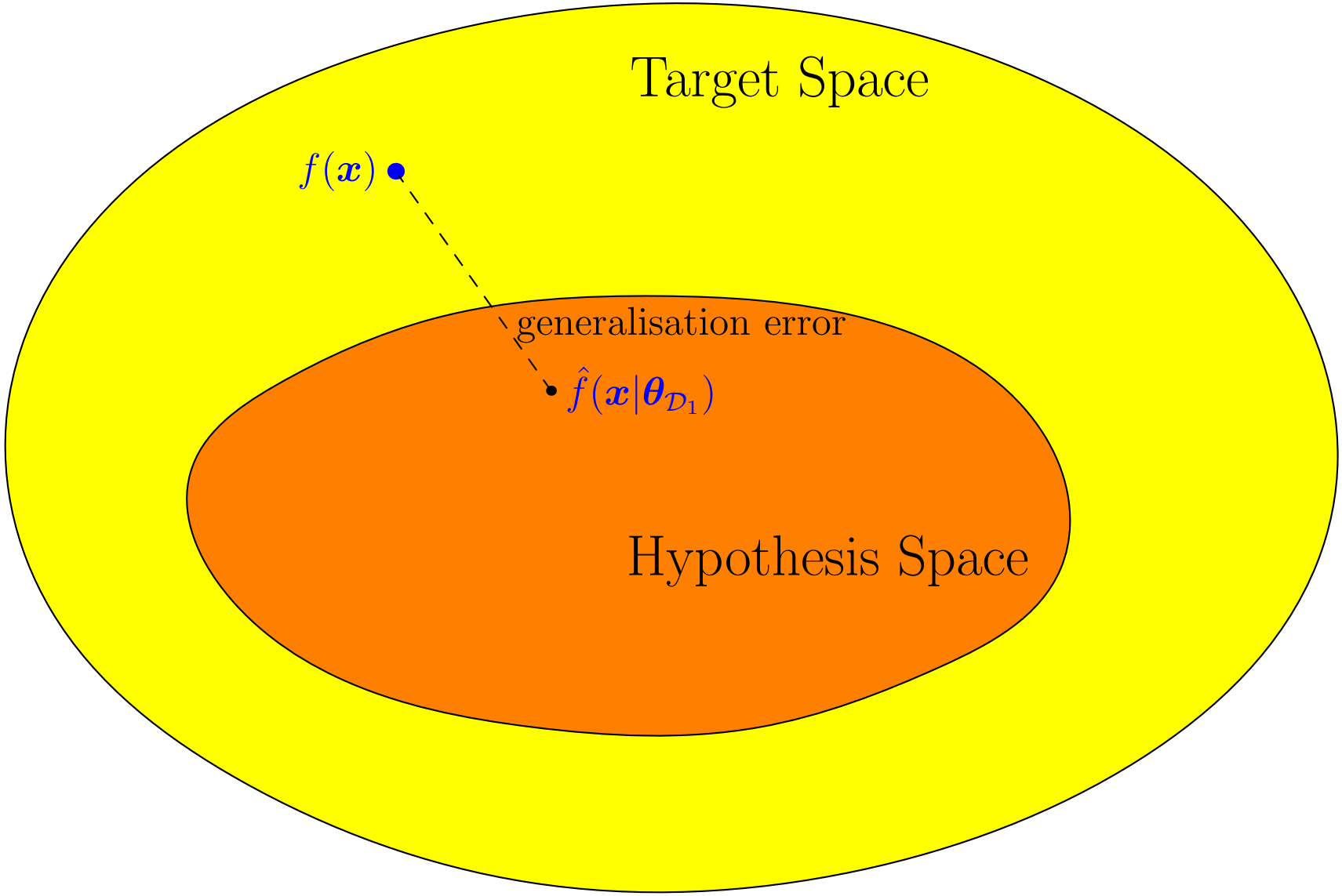
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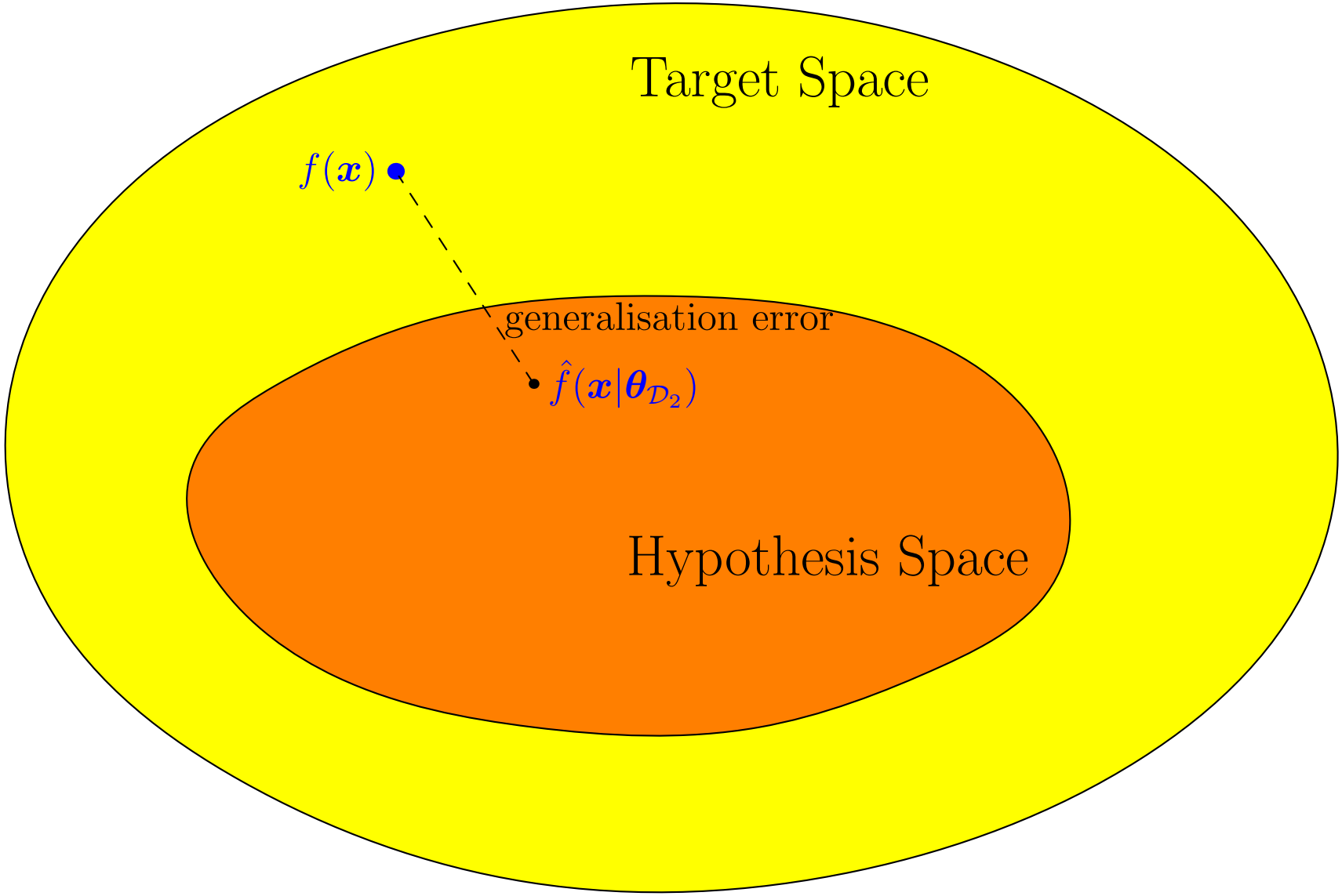
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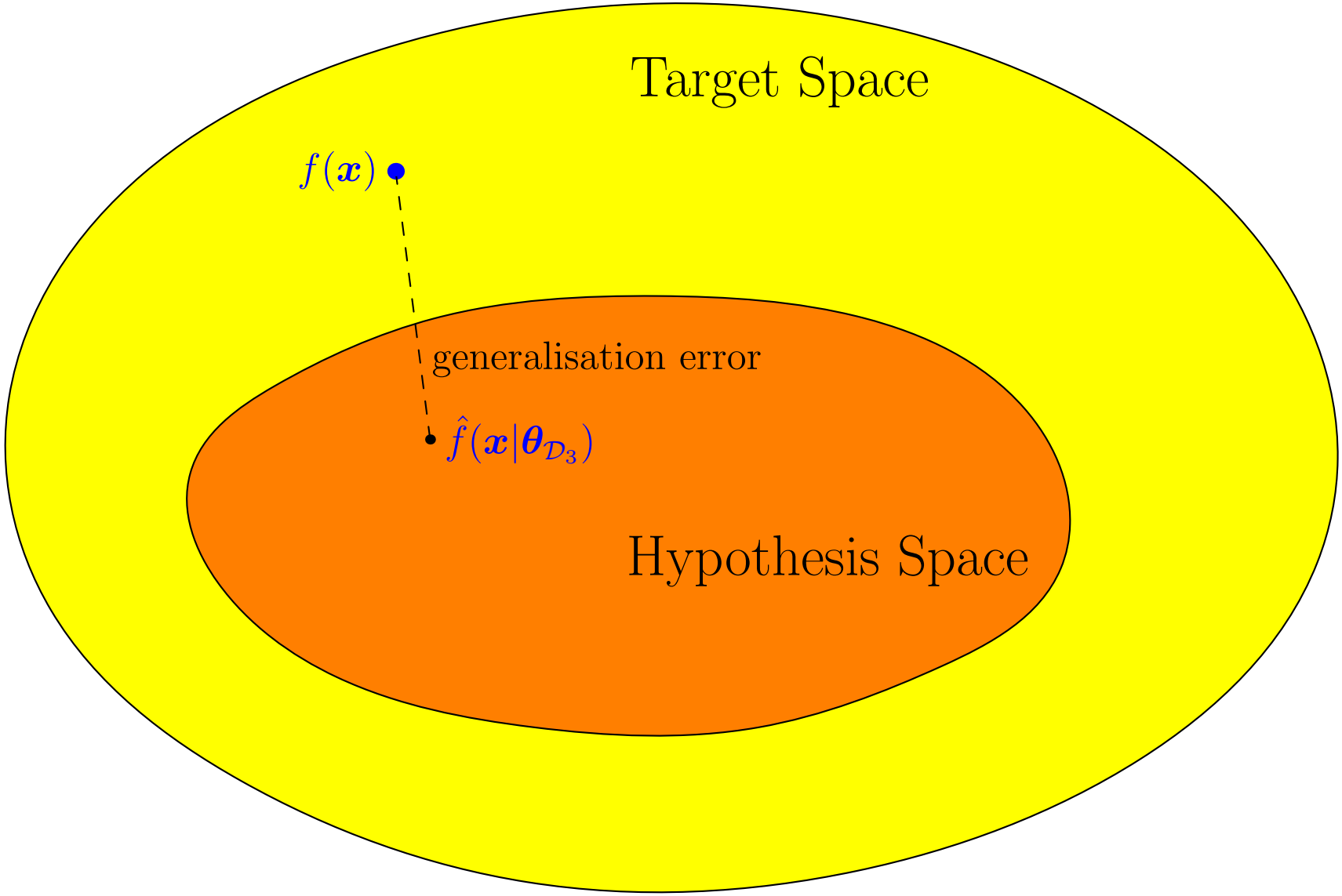
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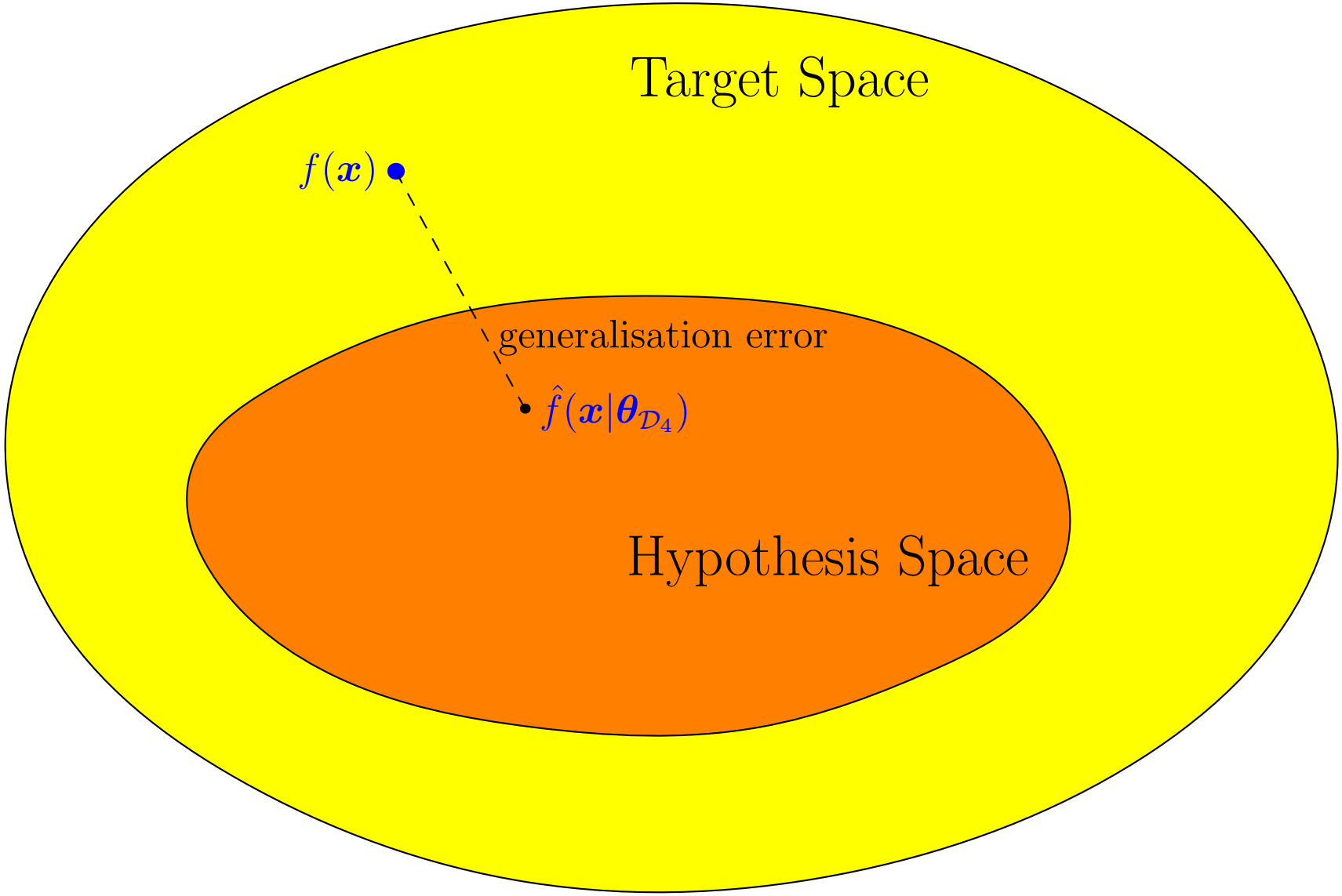
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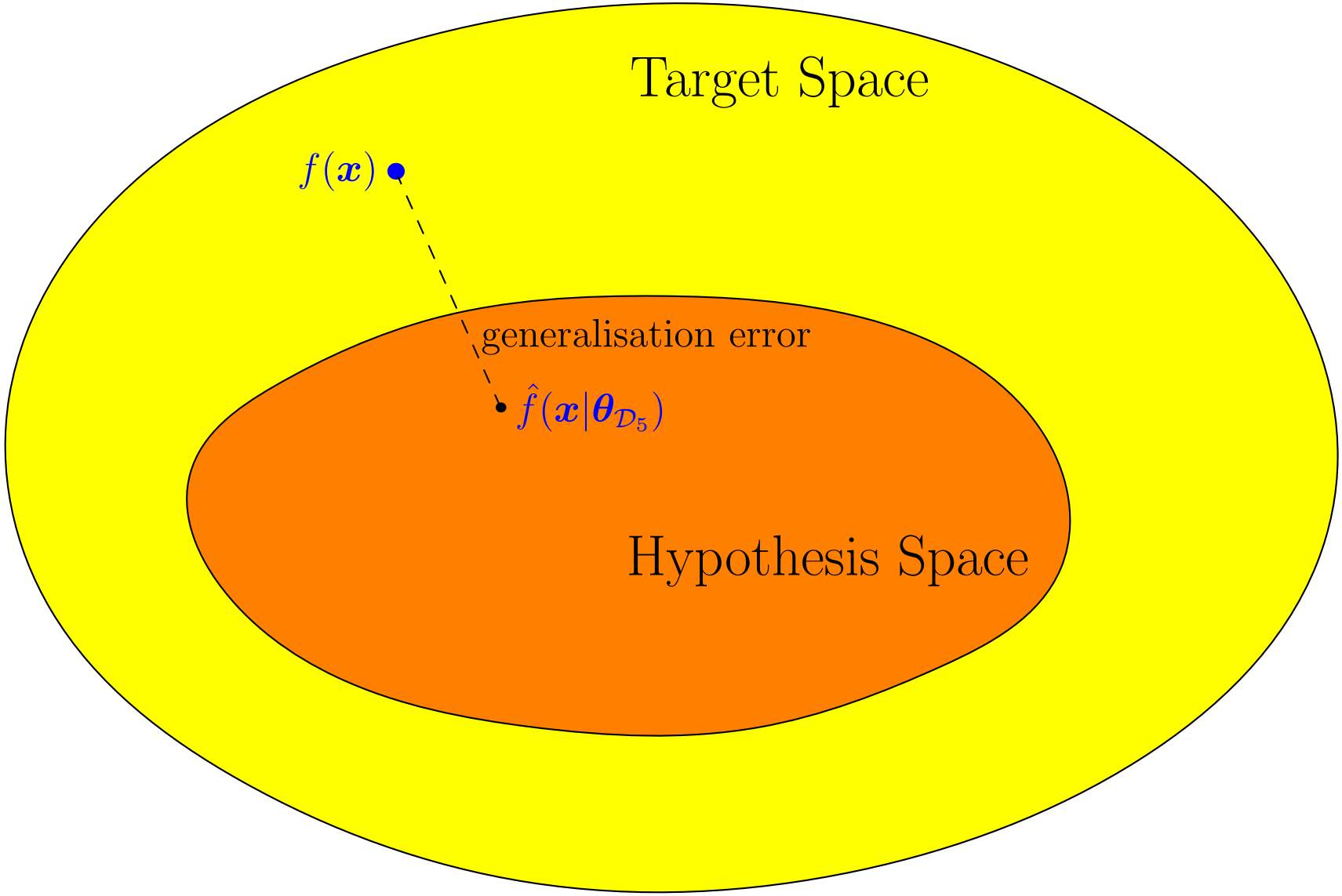
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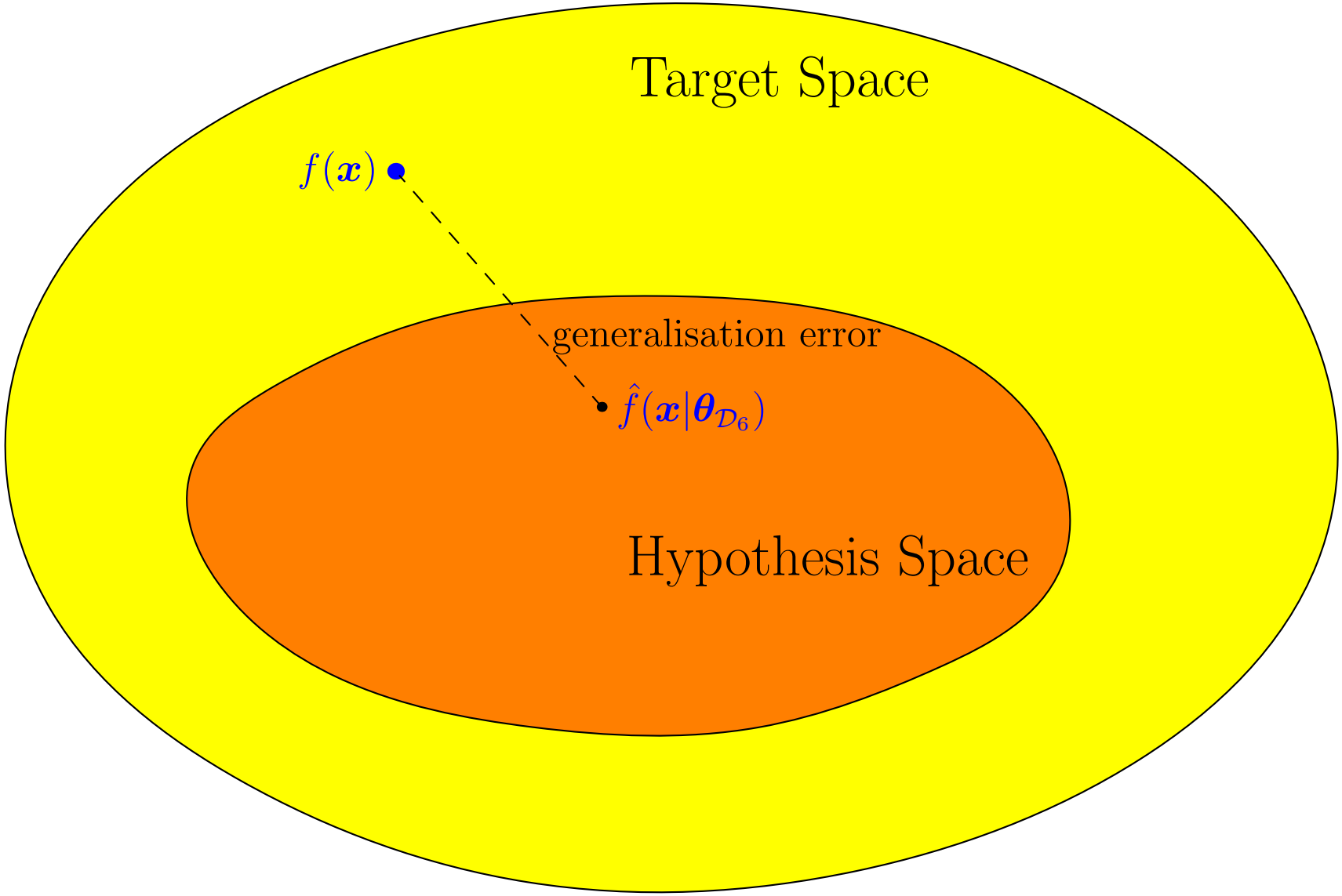
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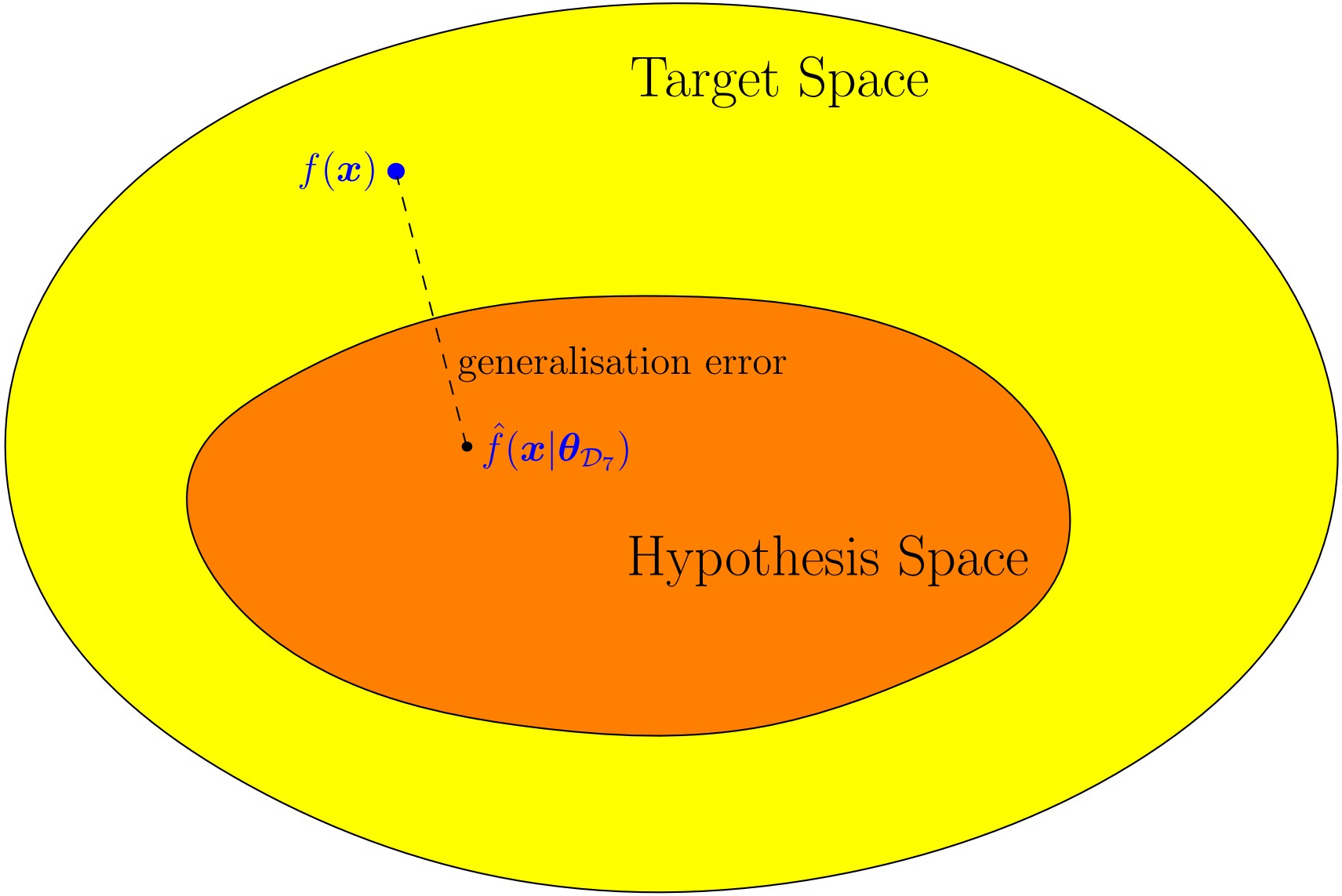
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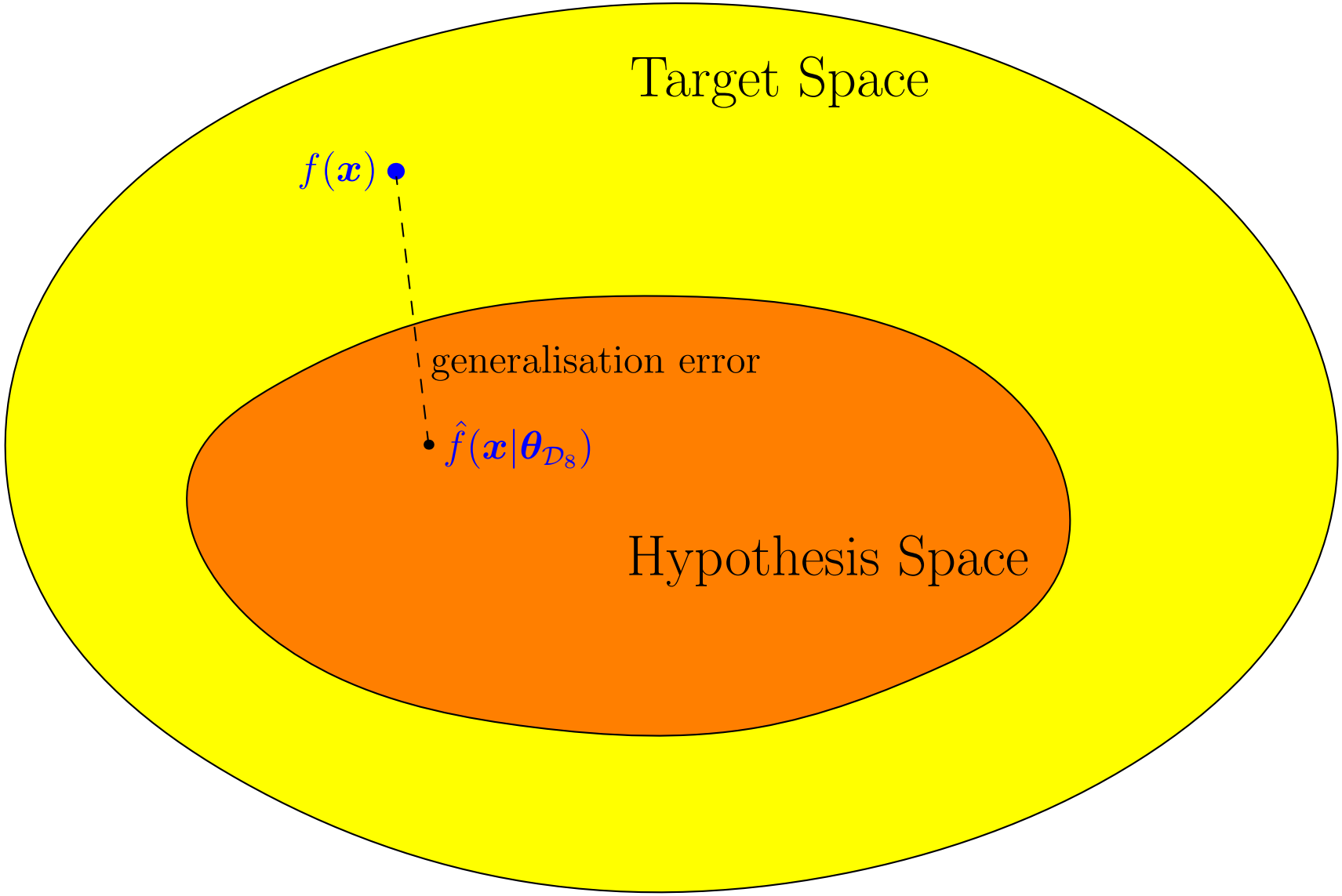
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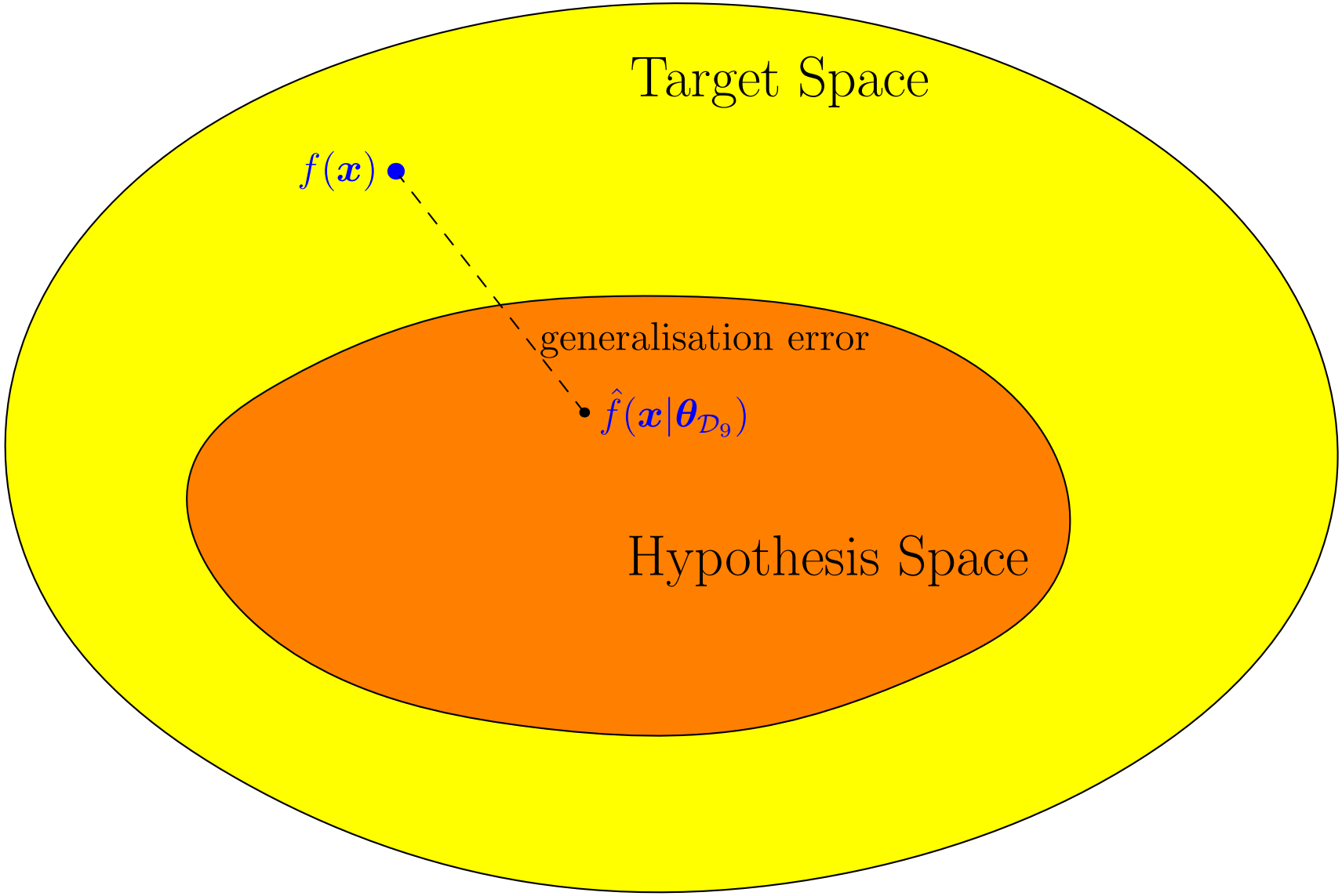
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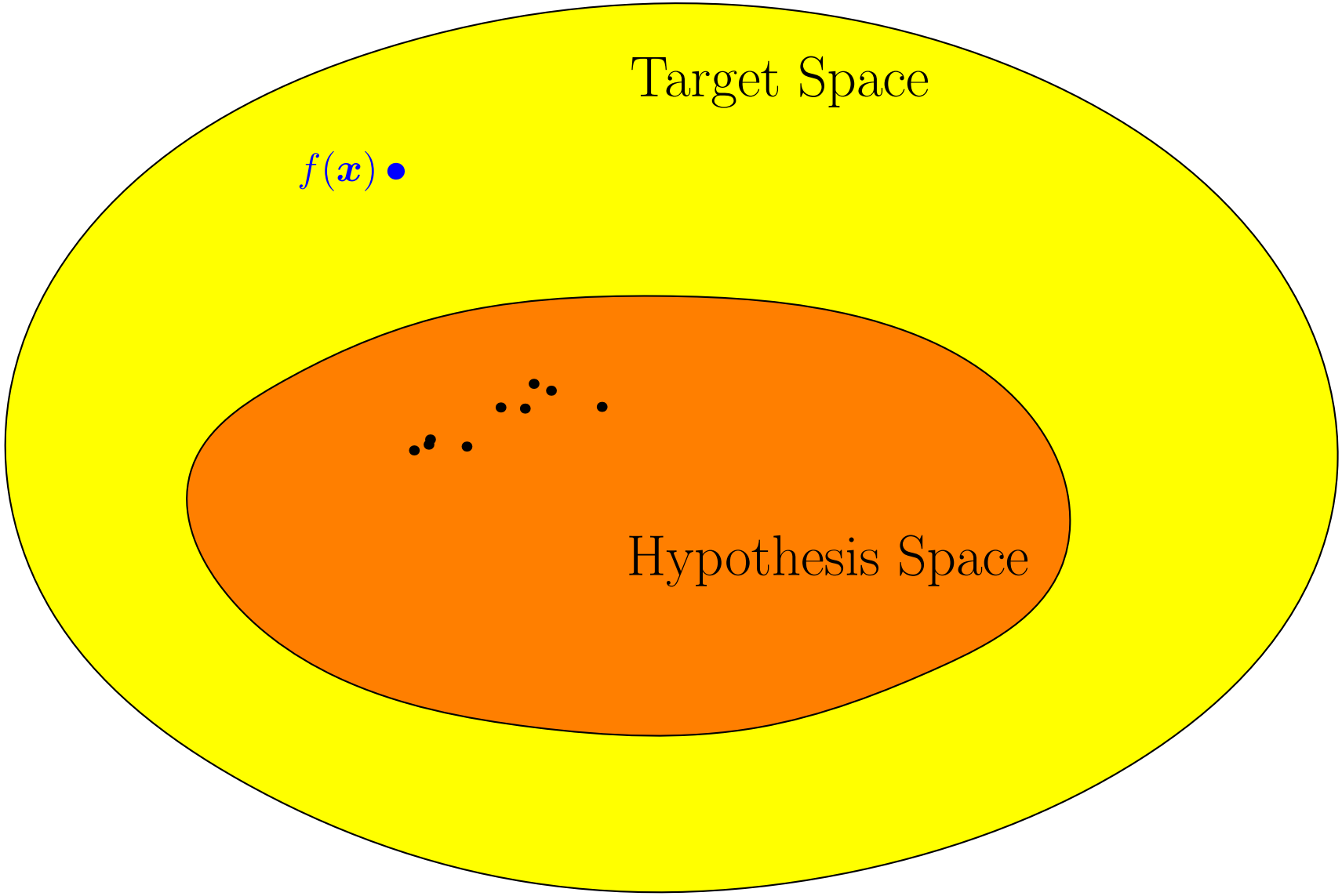
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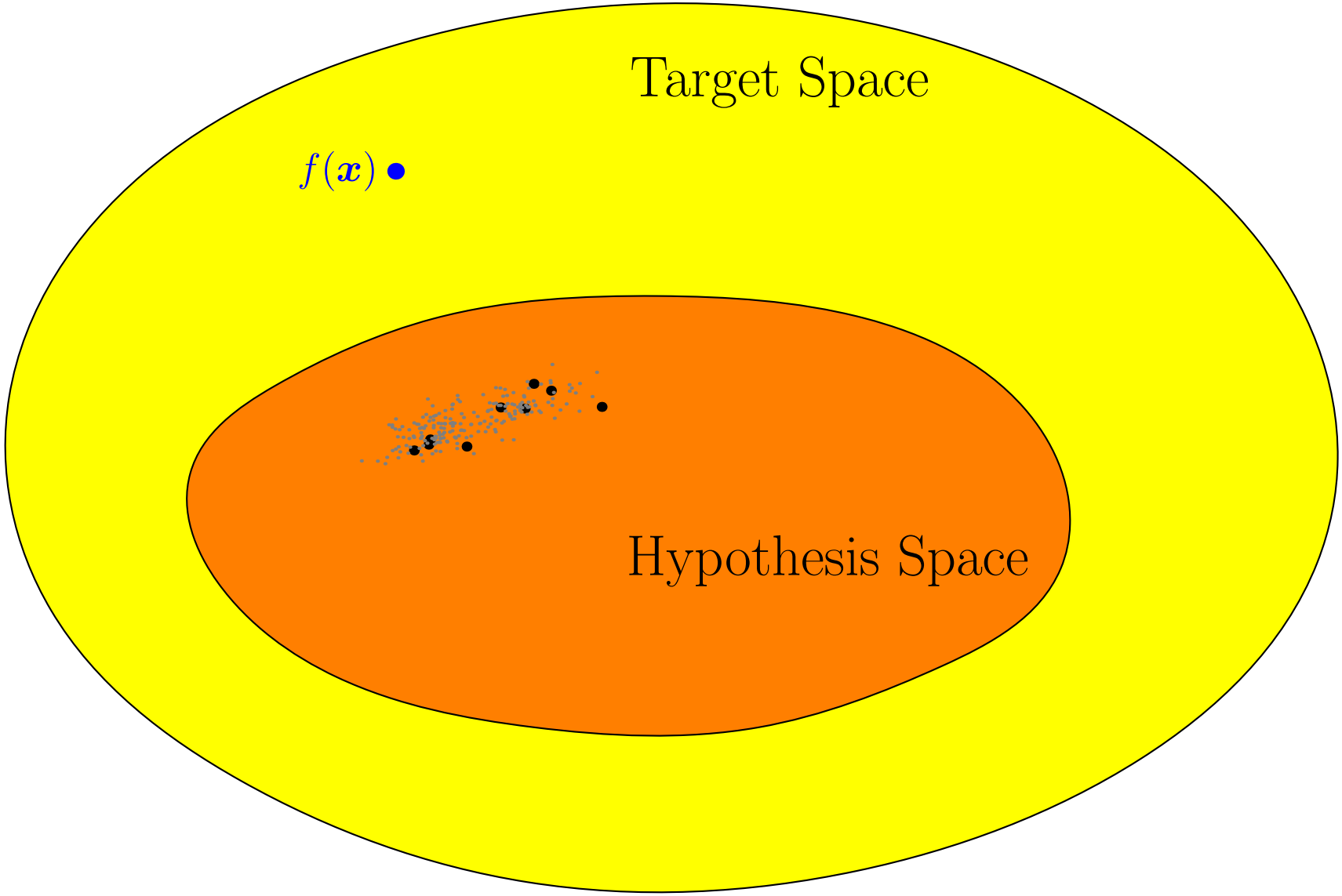
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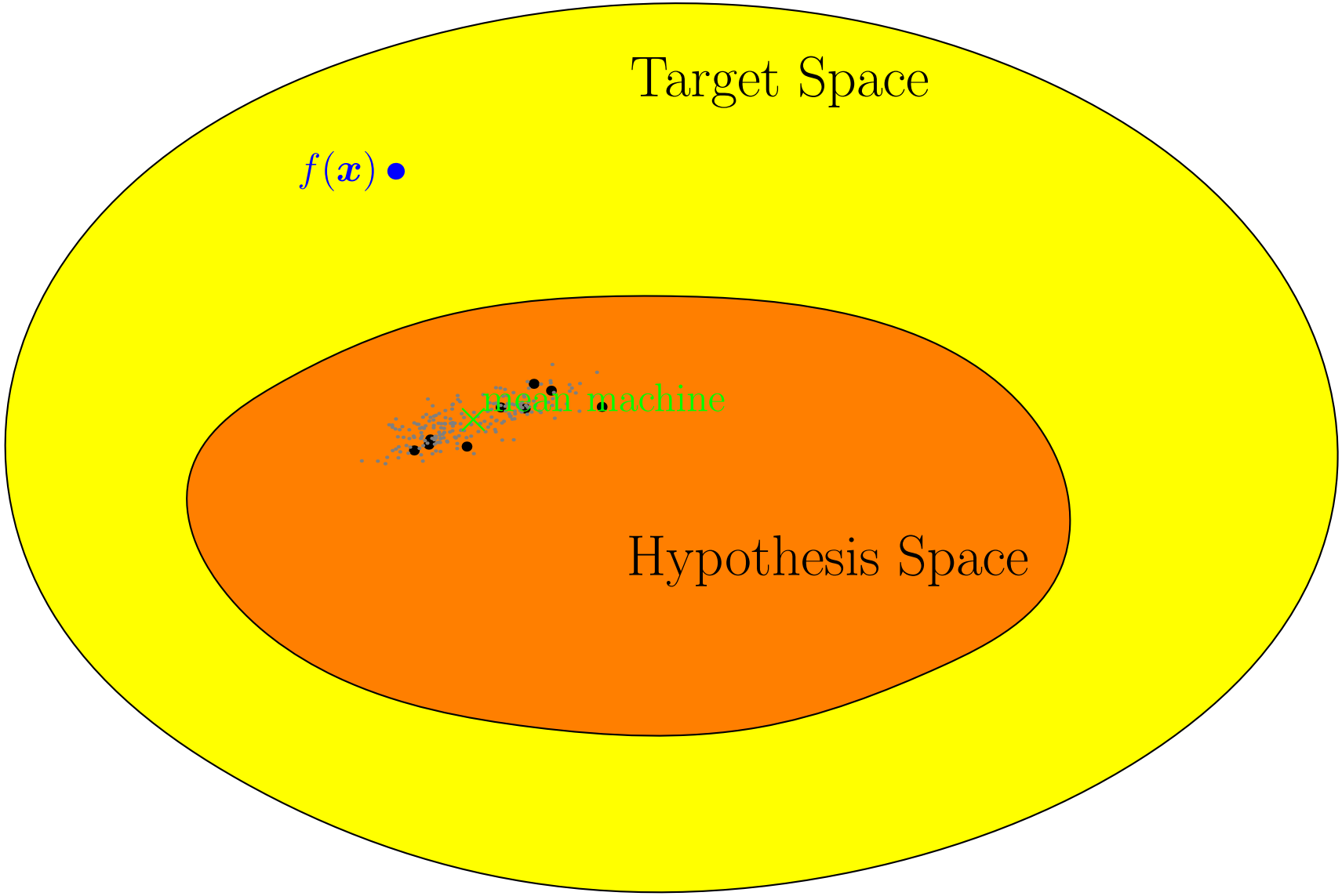
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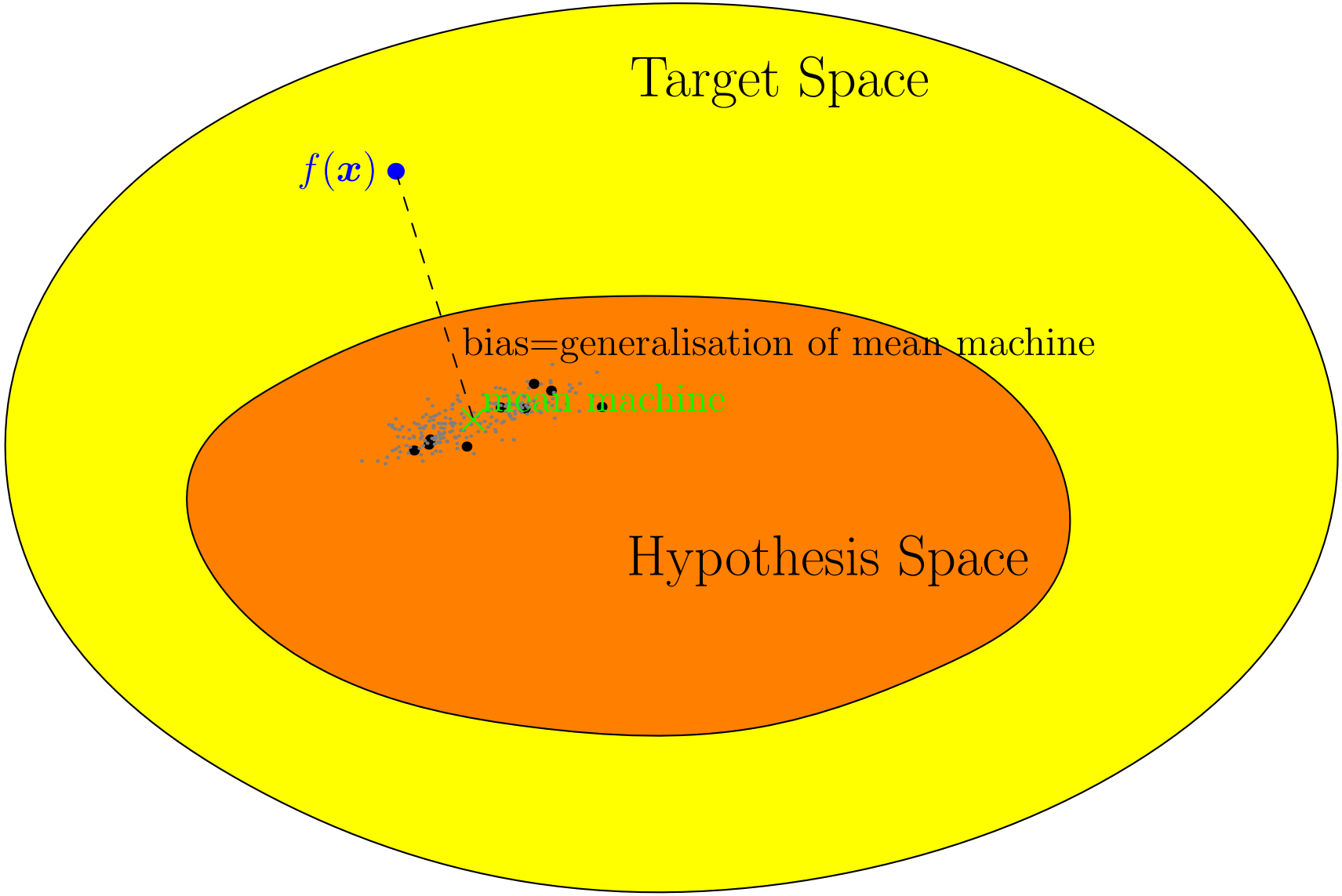
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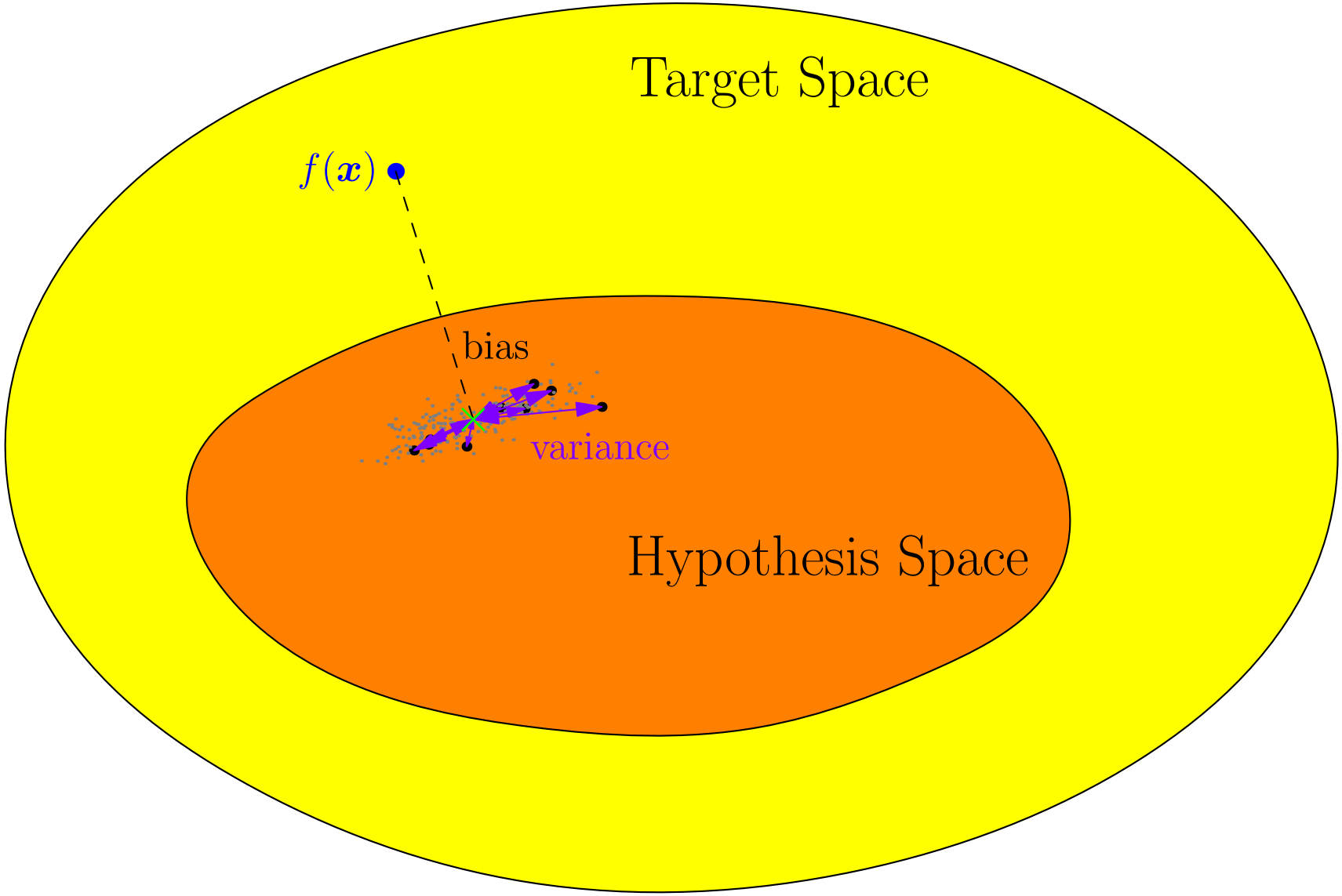
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Approximation and Estimation Errors



Mean Machine

- To help understand generalisation we can consider the mean prediction with respect to machines trained with all data sets of size m

$$\hat{f}_m(\boldsymbol{x}) = \mathbb{E}_{\mathcal{D}}\left[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})\right]$$

- We can define the **bias** to be generalisation performance of the mean machine

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\hat{f}_m(\boldsymbol{x}) - y\right)^2$$

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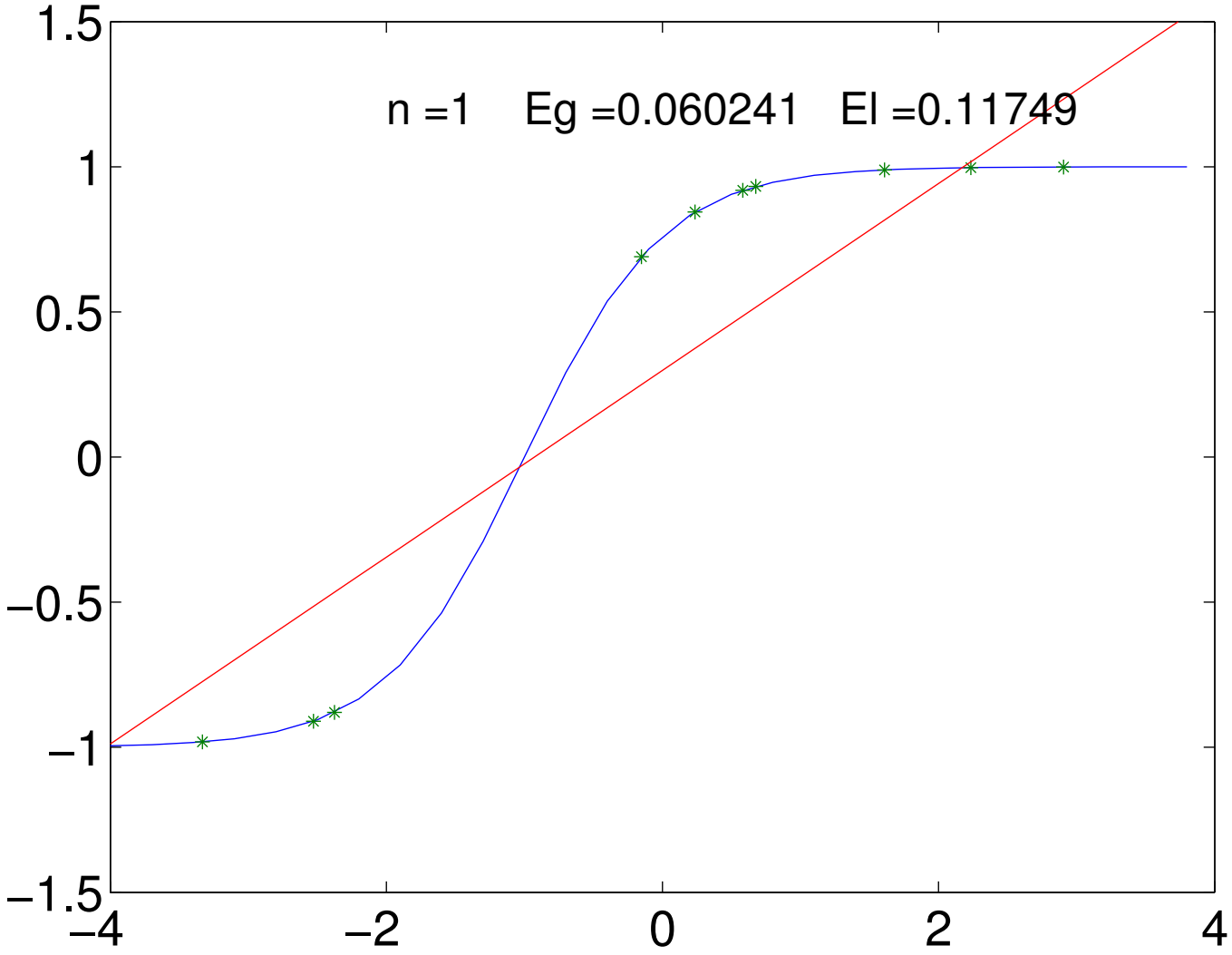
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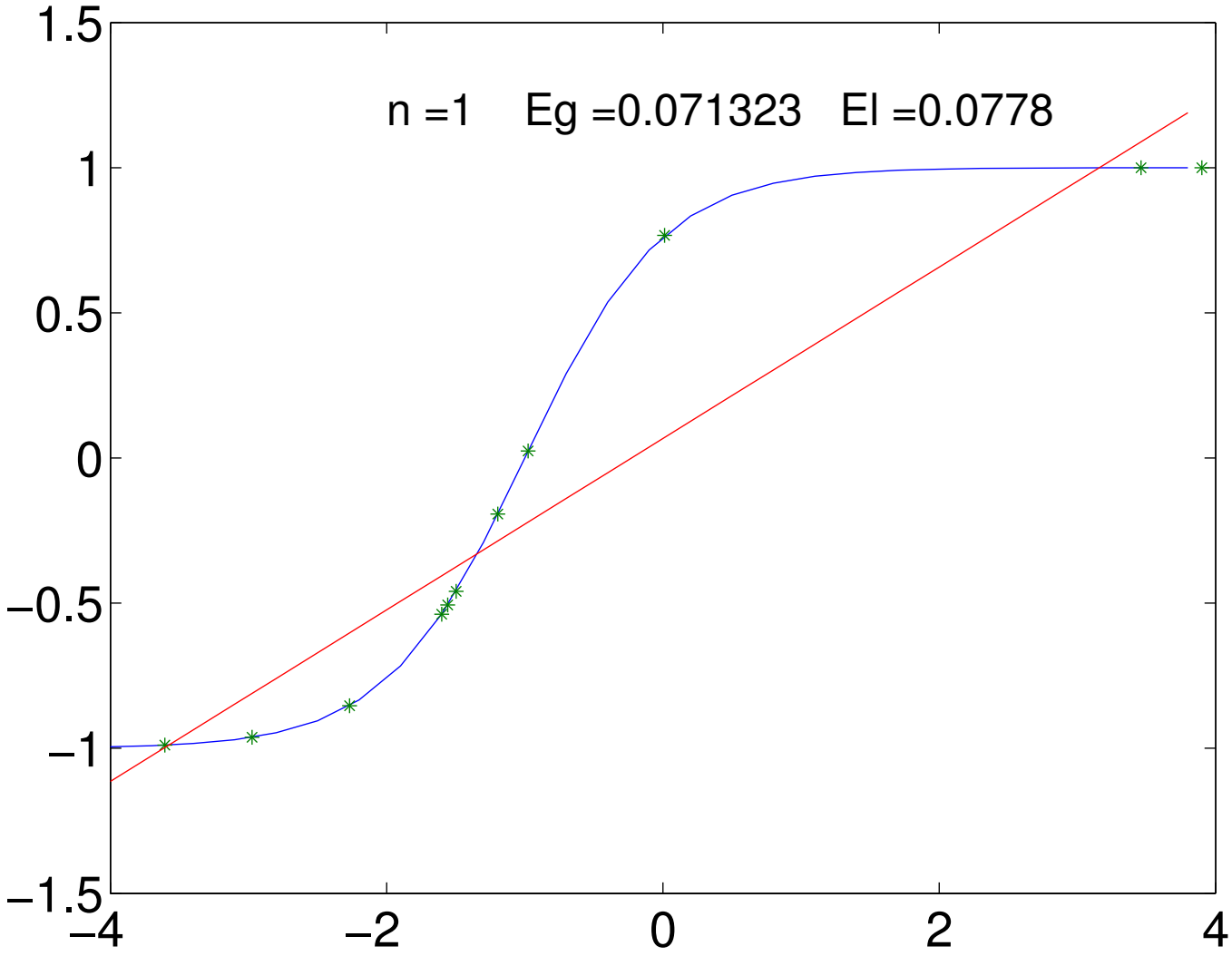
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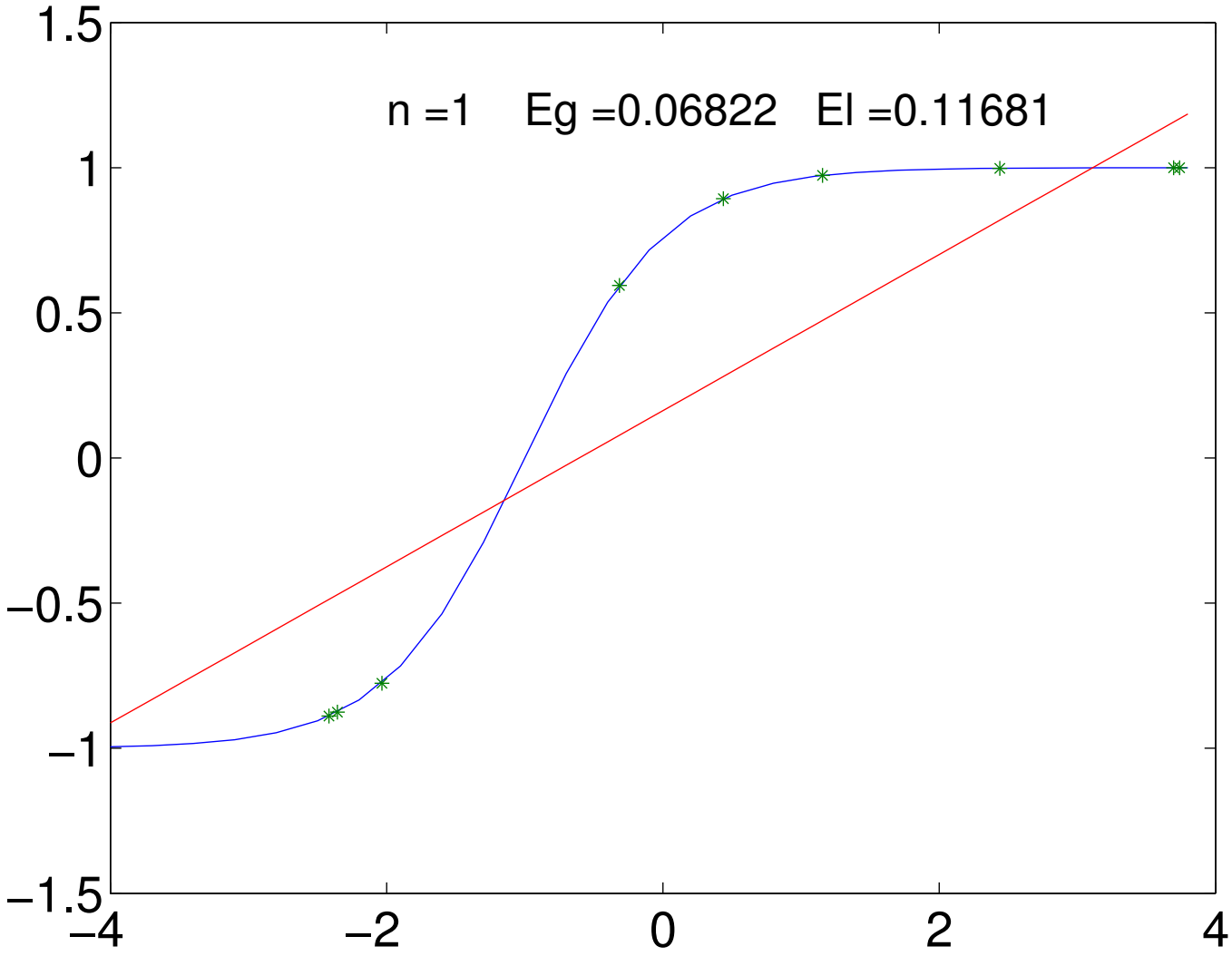
Regression Example $n = 1$



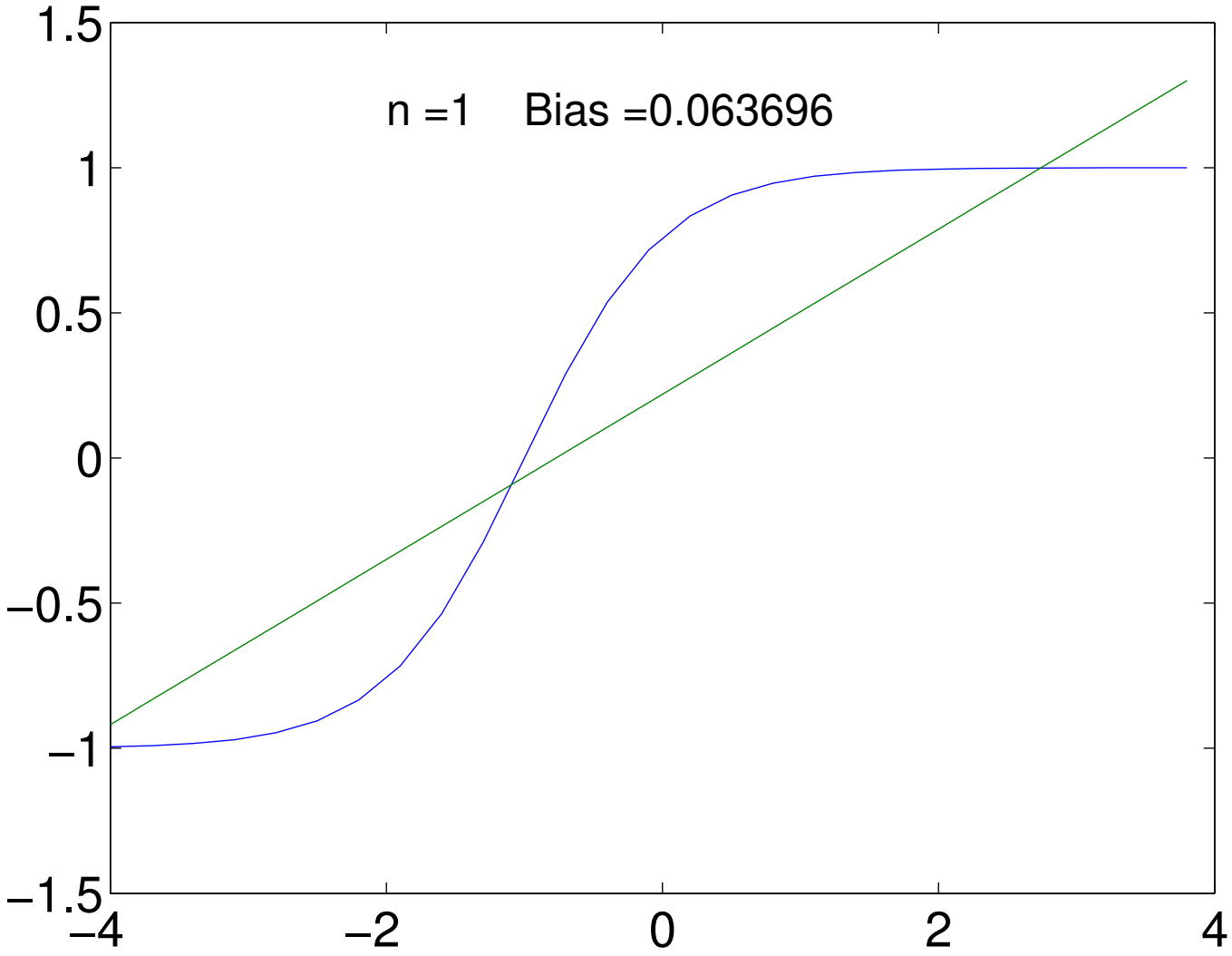
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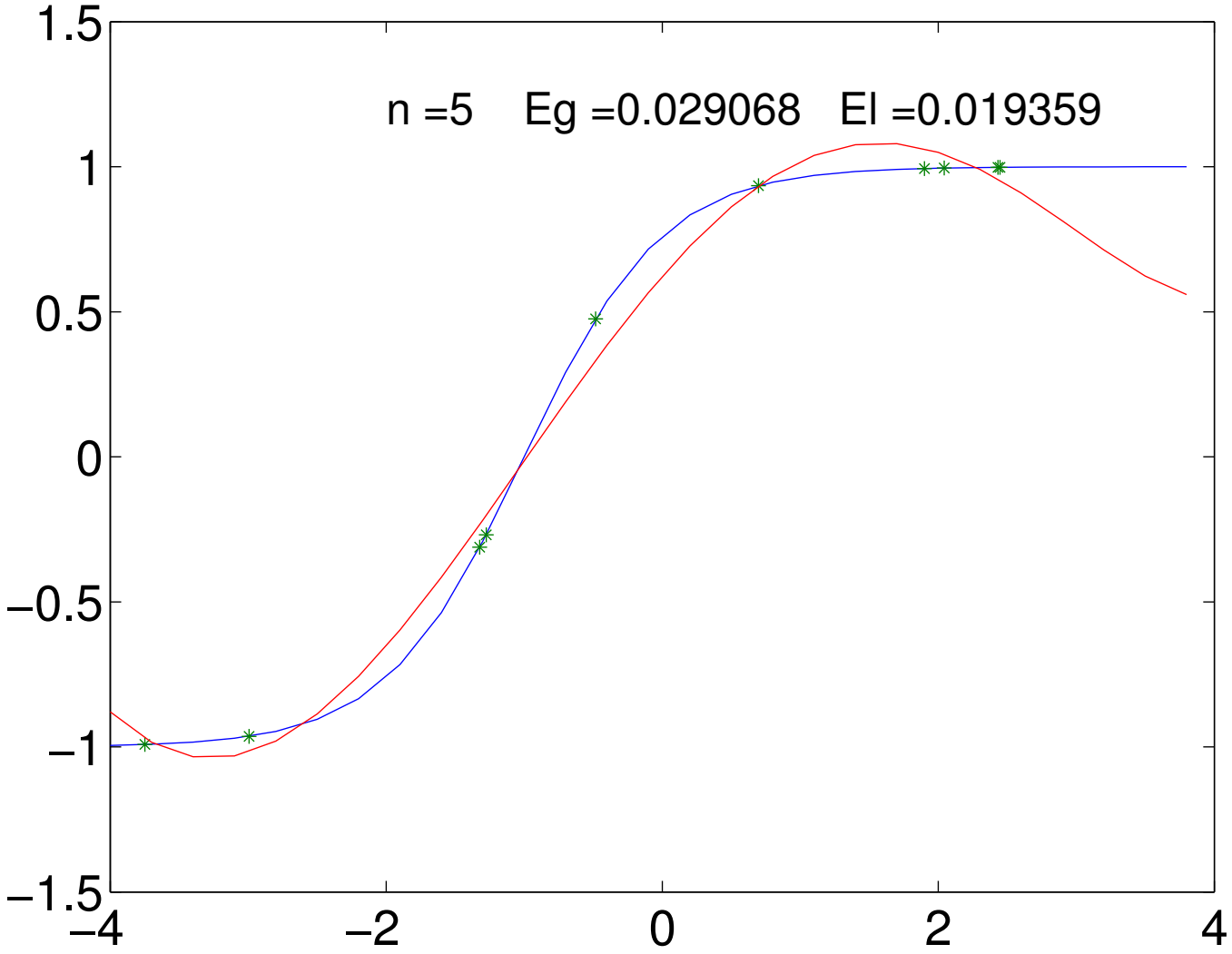
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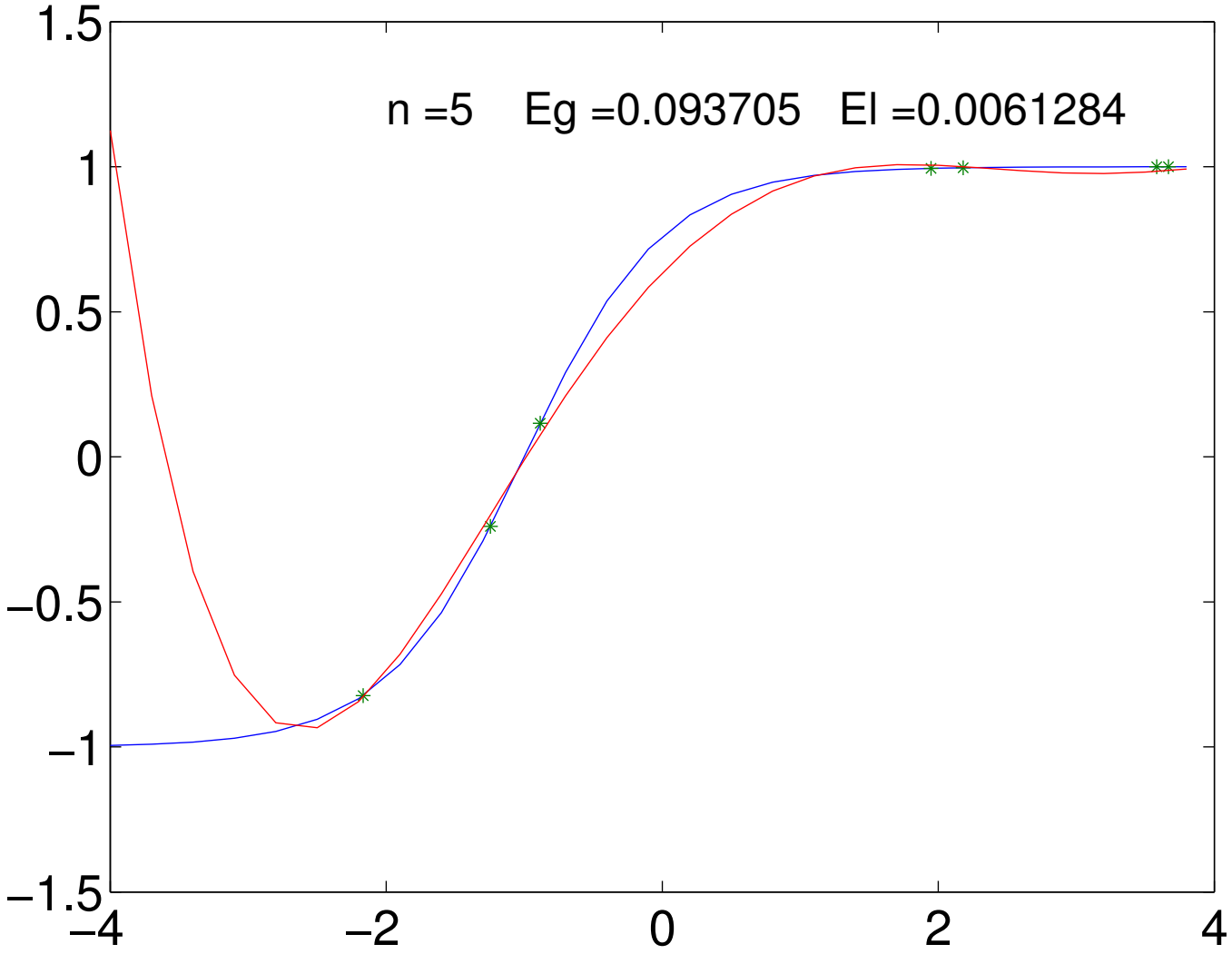
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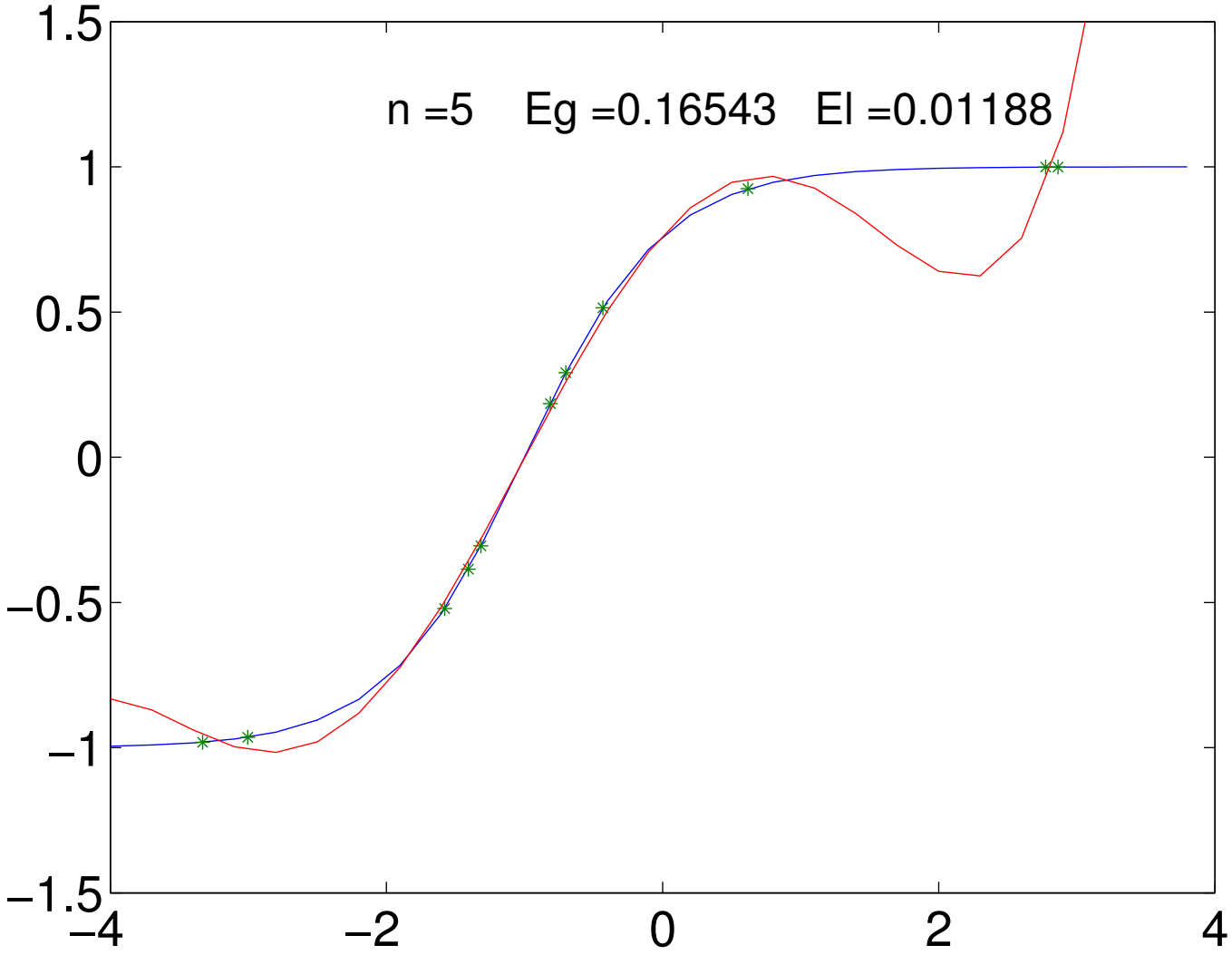
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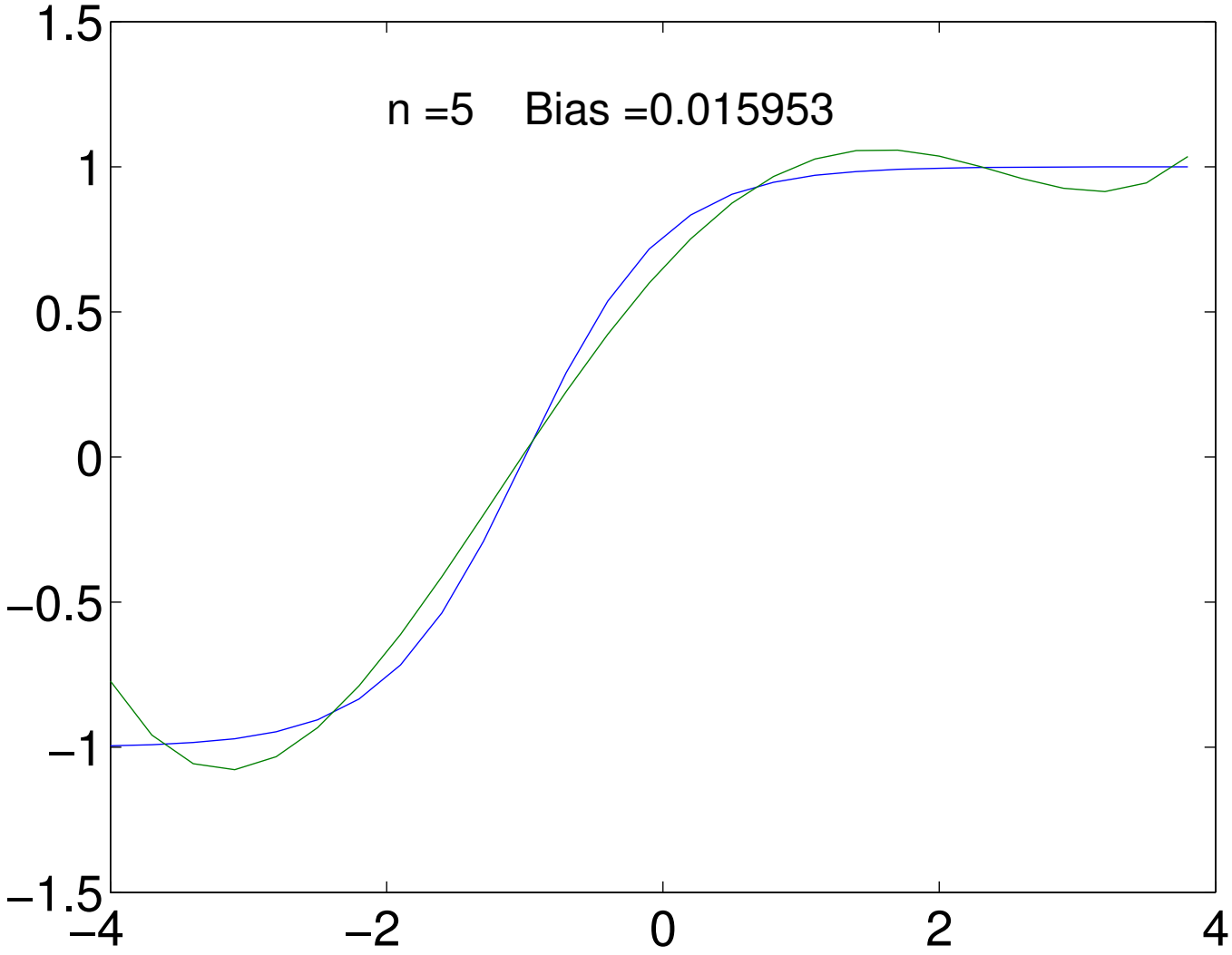
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- The cross term vanishes

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Bias and Variance

- We can write the expected generalisation loss as

$$\mathbb{E}_{\mathcal{D}}[L_G(\boldsymbol{\theta}_{\mathcal{D}})] = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2\right] + \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\hat{f}_m(\boldsymbol{x}) - y\right)^2 = V + B$$

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Where Is the Noise?

- Suppose $y = f^*(\boldsymbol{x}) + \epsilon$ where the noise ϵ has mean 0 and variance σ^2
- Then (the cross term vanishes again, as $\mathbb{E}[\epsilon] = 0$)

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\big(\hat{f}_m(\boldsymbol{x}) - y\big)^2 = \sigma^2 + \sum_{\boldsymbol{x}} \mu(\boldsymbol{x})\big(\hat{f}_m(\boldsymbol{x}) - f^*(\boldsymbol{x})\big)^2$$

- So $\bar{L}_G = \text{noise} + \text{bias}^2 + \text{variance}$, the form you may have seen elsewhere
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Bias-Variance Dilemma

- The bias measures the generalisation performance of the *mean machine* and is large if the machine is too simple to capture the changes in the function we want to learn
- The variance measures the variation in the prediction of the machines as we change the data set we train on

$$V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y) \mathbb{E}_{\mathcal{D}} \left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x}) \right)^2 \right]$$

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- We want to choose a learning machine that is complex enough to capture the underlying function we are trying to learn, but otherwise as simple as possible
- There are a number of tricks to achieve this balance
- Some require us to preprocess the data to reduce the number of inputs
- Some machines cleverly adjust their own complexity
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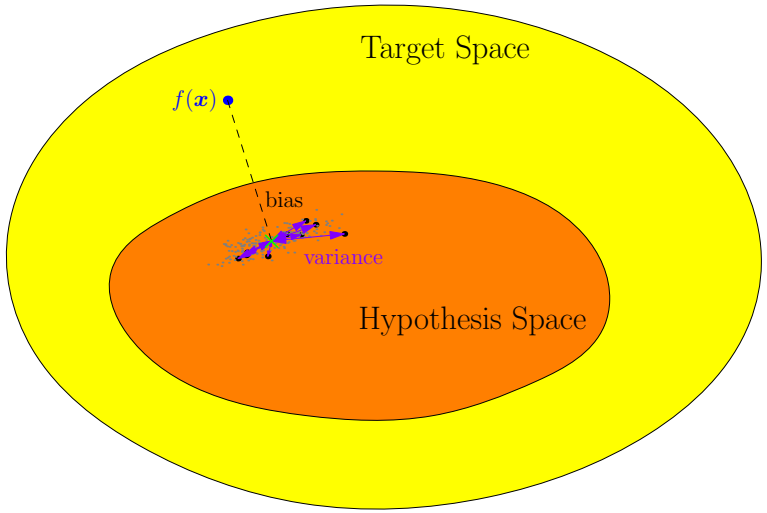
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Outline

1. What Makes a Good Learning Machine?
2. Bias-Variance Dilemma
3. **Shrinkage and Ridge Regression**
4. Ensembles



Shrinking an Estimate

- The simplest learning problem: estimate the mean θ of a distribution with variance σ^2 from m samples
- The sample mean $\bar{x}$ is unbiased with variance σ^2/m
- Try the deliberately biased $\hat{\theta} = c\bar{x}$ with $0 \leq c \leq 1$

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Ridge Regression

- Data $\boldsymbol{y} = \mathbf{X}\boldsymbol{w}^* + \boldsymbol{\epsilon}$, with noise of variance σ^2 on each target
- Ridge regression minimises $\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \eta\|\boldsymbol{w}\|^2$, giving

$$\hat{\boldsymbol{w}} = \left(\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\mathbf{I}\right)^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{y}$$

- $\eta = 0$ is least squares: unbiased, but how variable?
- Use the SVD $\mathbf{X} = \mathbf{U}\boldsymbol{S}\boldsymbol{V}^\mathsf{T}$ and work in the basis of the columns of $\boldsymbol{V}$, where everything separates

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Ridge Through the SVD

- Write $w_k^* = \boldsymbol{v}_k^\mathsf{T} \boldsymbol{w}^*$ for the component of $\boldsymbol{w}^*$ along $\boldsymbol{v}_k$; since $\boldsymbol{u}_k^\mathsf{T} \boldsymbol{y} = s_k w_k^* + \boldsymbol{u}_k^\mathsf{T} \boldsymbol{\epsilon}$, the same component of the estimate is

$$\hat{w}_k = \frac{s_k \boldsymbol{u}_k^\mathsf{T} \boldsymbol{y}}{s_k^2 + \eta}, \qquad \mathbb{E}[\hat{w}_k] = \frac{s_k^2}{s_k^2 + \eta} w_k^*, \qquad \mathrm{Var}(\hat{w}_k) = \frac{\sigma^2 s_k^2}{(s_k^2 + \eta)^2}$$

- So the expected squared error of the weights is

$$\mathbb{E}\left[\|\hat{\boldsymbol{w}} - \boldsymbol{w}^*\|^2\right] = \sum_k \underbrace{\left(\frac{\eta}{s_k^2 + \eta}\right)^2 w_k^{*2}}_{\text{bias}^2} + \underbrace{\frac{\sigma^2 s_k^2}{(s_k^2 + \eta)^2}}_{\text{variance}}$$

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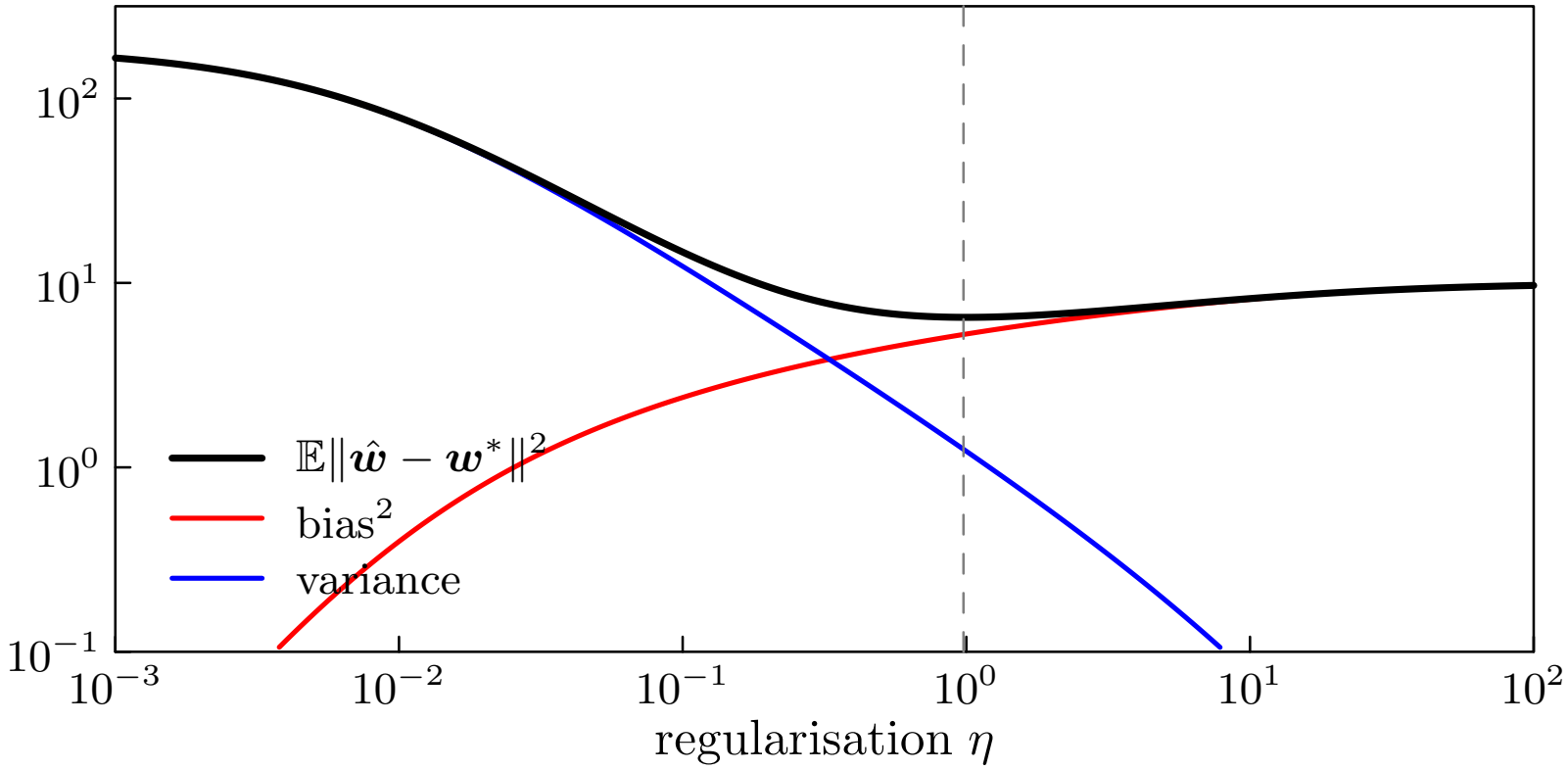
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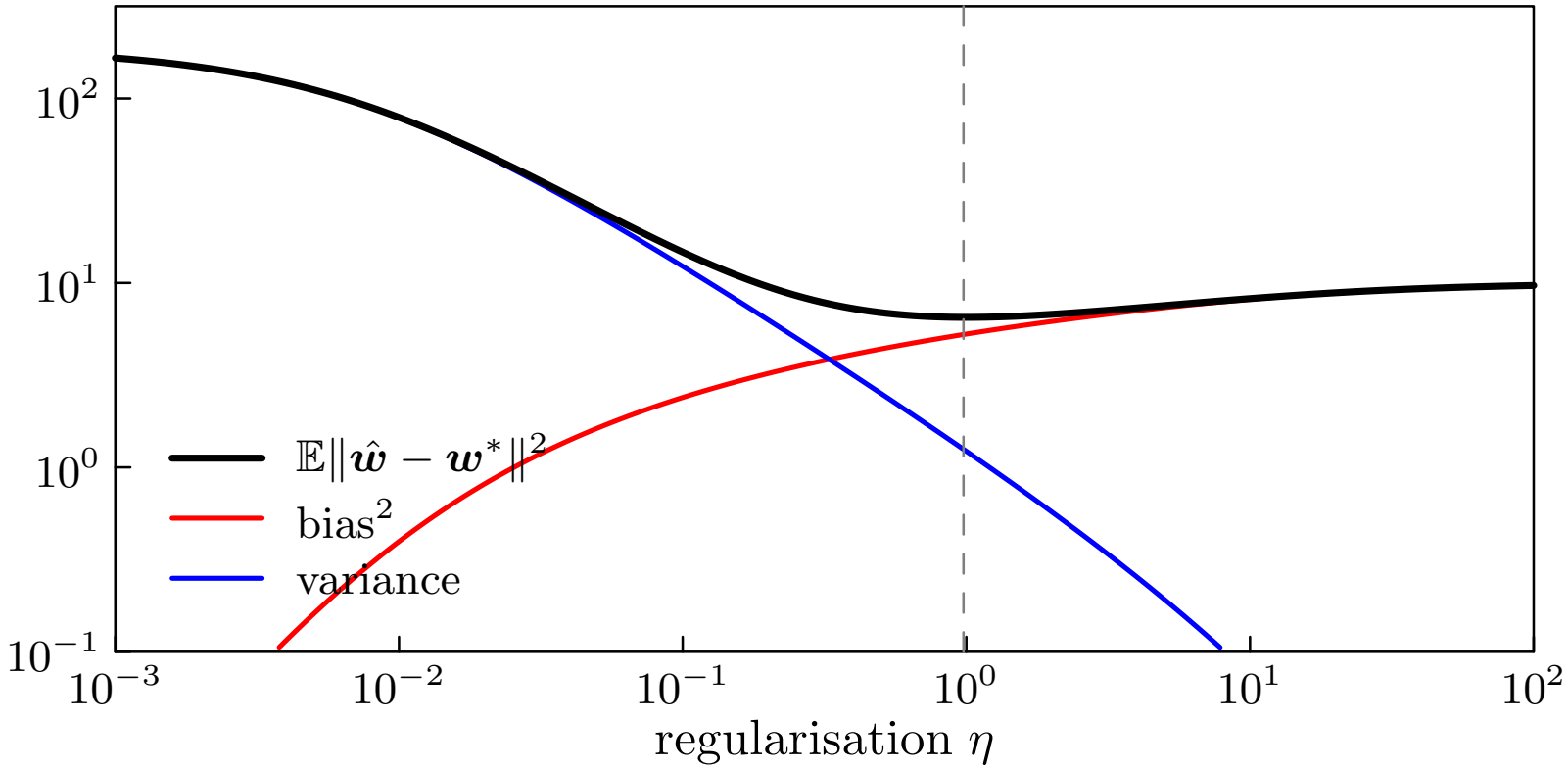
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Choosing η



- Ten singular values from 3 down to 0.1, $\sigma = 1$: least squares has error 188, the best $\eta \approx 1$ gives 6.5
- In practice choose η by cross-validation

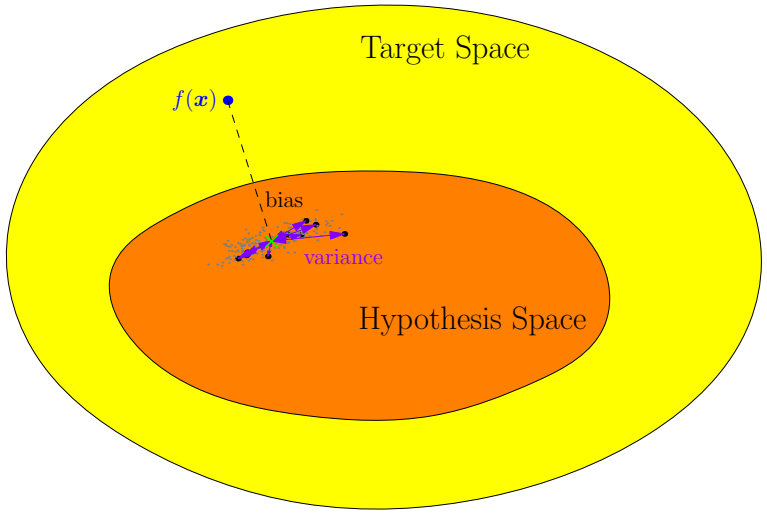
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Outline

1. What Makes a Good Learning Machine?
2. Bias-Variance Dilemma
3. Shrinkage and Ridge Regression
4. **Ensembles**



Averaging Reduces Variance

- An **ensemble** averages M predictors, $\bar{f}_M(\boldsymbol{x}) = \frac{1}{M}\sum_{i=1}^{M} f_i(\boldsymbol{x})$, trained with different randomness
- Averaging does not change the mean, so the bias is unchanged
- If each $f_i(\boldsymbol{x})$ has variance s^2 and each pair has correlation ρ

$$\mathrm{Var}\big(\bar{f}_M(\boldsymbol{x})\big) = \rho s^2 + \frac{1-\rho}{M} s^2$$

- More members remove the second term, but correlation sets a floor ρs^2
- So we want members that are good **and** different

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- **Bagging**: train each member on a bootstrap resample of the data (sampled with replacement)
- **Random forests**: bagged decision trees that also choose each split from a random subset of the features

- **Deep ensembles**: train several networks on the same data from different random initialisations
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- One of the most useful intuitions about what works is the bias-variance framework
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